NOT MEASUREMENT SENSITIVE

MIL-T-62110G
16 April 1990
SUPERSEDING
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MILITARY SPECIFICATION

TRUCKS, MAINTENANCE: WITH ROTATING HYDRAULIC DERRICK, 32,000 TO 44,500 GVW, 4x2, 4x4, 6x4 AND 6x6, COMMERCIAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers maintenance trucks with a rotating and telescoping hydraulic derrick. Vehicles procured under this specification are commercial items which are warranted by the manufacturer as specified in acquisition documents.

1.2 Classification. The vehicle shall be one of the types, models, classes, and sizes shown in table I, as specified (see 6.2). The model of the vehicle shall be determined by the number of wheels and number of driving wheels. The size of the vehicle shall be the length of the pole that the derrick shall be capable of setting.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-UED, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC-2320

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TABLE I. Vehicle classification.

Type	Derrick location and body type	Model (wheels and wheels driving)	Class	Size	Pole length (feet)
I	Rear of cab, no racks above cabinets, no roof	4x2, 4x4	A	55 70	55 70
		6x4	B	70	70
		6x6	C	70	70
II	Rear corner, racks above cabinets and telescoping roof	4x2, 4x4	A	55 70	55 70
*		6x4	B D	70	70
III	Center rear, over rear axle(s), no racks above cabinets, no roof	4x2, 4x4	A	55 70	55 70
		6x4	B	70	70
		6x6	C	70	70

1.2.1 Class. The class of a vehicle shall be determined by the minimum gross vehicle weight (GVW) rating as specified in table II:

TABLE II. GVW ratings.

Vehicle class	GVW rating (pounds)
* A	32,000
B	34,000
C	36,000
* D	44,500

* 2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS), and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

W-B-131 - Battery, Storage: Vehicular, Ignition, Lighting and Starting.

* VV-F-800 - Fuel Oil, Diesel

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MILITARY

	MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
*	MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4 and JP-5.
*	MIL-T-83133	- Turbine Fuel, Aviation, Kerosene Type, Grade JP-8.
	MIL-H-83282	- Hydraulic Fluid, Fire-Resistant, Synthetic Hydrocarbon Base, Aircraft, NATO Code Number H-537.

STANDARDS

FEDERAL

FED-STD-297	- Rustproofing of Commercial (Nontactical) Vehicles.
FED-STD-595	- Colors.

MILITARY

MIL-STD-1223	- Nontactical Wheeled Vehicles Treatment, Painting, Identification Marking and Data Plate Standards.
MS51118	- Pintle Assembly, Towing - 40,000 Lbs. Capacity, Manual Release.
MS51317	- Light, Warning, Vehicular, Rotating, D.C.

* (Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

* DEPARTMENT OF DEFENSE

Department of Defense Index of Specifications and Standards (DODISS).

(Copies of the DODISS are available on a yearly subscription basis either form the Government Printing Office for hard copy, or microfiche copies are available from the Director, Navy Publication and Printing Service Office, 700 Robbins Avenue, Philadelphia, PA 9111-5093.)

DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Carrier Safety Regulations.
Federal Motor Vehicle Safety Standards.

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(Application for copies of DoT publications should reference the Code of Federal Regulations, 49 CFR, and the Federal Register and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)
Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.
Noise Emission Standards for Transportation Equipment - Medium and Heavy Trucks.

(Application for copies of EPA publications should reference the Code of Federal Regulations, 40 CFR, and the Federal Register and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)
Occupational Safety and Health Standards.
1926.556 - Aerial Lifts.

(Application for copies of OSHA publications should reference the Code of Federal Regulations, Title 29, and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

* 2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issue of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
ANSI A92.2 - American National Standard for Vehicle-Mounted Elevating and Rotating Aerial Devices.

(Application for copies of ANSI publications should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
ASTM Standards.
A120 - Pipe, Steel, Black and Hot-Dipped Zinc-Coated (Galvanized) Welded and Seamless, for Ordinary Uses (DoD adopted).
B117 - Standard Method of Salt Spray (Fog) Testing.

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(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

* THE EUROPEAN TYRE AND RIM TECHNICAL ORGANISATION (ETRTO)
Standards Manual

(Application for copies of the ETRTO publications should be addressed to the European Tyre and Rim Technical Organisation, 32, Avenue Brugmann, 1060 Brussels, Belgium.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)
NFPA-70 - National Electrical Code.

* NORTH ATLANTIC TREATY ORGANIZATION (NATO)
Allied Engineering Publication (AEP-5)
Part II - NATO Standard Diesel and Spark Ignition Engines Laboratory Test, dated January 1984.

(Application for copies of NATO publications should be addressed to North Atlantic Treaty Organization, Military Agency for Standardization, 1110 Brussels, Belgium.)

SAE, INC.
SAE Standards and Recommended Practice.

J318	- Air Brake Gladhand Service (Control) and Emergency (Supply) Line Couplers - Trucks, Truck-Tractors, and Trailers (DoD adopted).
J350	- Spark Arrester Test Procedure for Medium Size Engines (DoD adopted).
J537	- Storage Batteries.
J551	- Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (30-1000 MHz).
J560	- Seven-Conductor Electrical Connector for Truck-Trailer Jumper Cable.
J588	- Turn Signal Lamps.
J589	- Turn Signal Switch.
J682	- Rear Wheel Splash and Stone Throw Protection (DoD adopted).
J683	- Tire Chain Clearance - Trucks, Buses, and Combinations of Vehicles.
J688	- Truck Ability Prediction Procedure (DoD adopted).
J704	- Openings for Six- and Eight-Bolt Truck Transmission Mounted Power Take-Offs.

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J706	-	Rating of Winches.
J844	-	Nonmetallic Air Brake System Tubing (DoD adopted).
J1067	-	Seven-Conductor Jacketed Cable for Truck-Trailer Connections.
J1292	-	Automobile, Truck, Truck-Tractor, Trailer and Motor Coach Wiring.
J1349	-	Engine Power Test Code - Spark Ignition and Diesel.

(Application for copies of SAE publications should be addressed to SAE, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION, INC.
Year Book.

(Application for copies of Tire and Rim Association publications should be addressed to The Tire and Rim Association, Inc., 3200 W. Market Street, Akron, OH 44313.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 Standard vehicle and accessories. Except as specified in 3.1.1 through 3.1.1.13, the vehicle, components, assemblies and accessories to be delivered under the contract shall be standard or optional items, which meet or exceed the requirements of this specification. Except as specified in 3.1.1 through 3.1.1.13, no removal, substitution or alteration of the chassis manufacturer's standard or optional chassis model components shall be made. All chassis items shall be as represented and rated in the chassis manufacturer's technical data book. The body, mounted derrick and equipment shall be as represented and rated in the body and equipment manufacturer's technical data. Technical data shall be limited to specifications and technical material identical to that furnished to the authorized company representatives for selection of vehicle models and components, and shall be available to the engineering offices of the procuring activity, prior to delivery of the items. The chassis model furnished shall be not older than the chassis manufacturer's current model on the date of the invitation for bids.

3.1.1 Special requirements. In addition to the standard vehicle and components specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.

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* 3.1.1.1 Treatment and painting. The vehicle shall be treated and painted in accordance with MIL-STD-1223. The exterior color shall be gloss yellow, matching color chip No. 13538 of FED-STD-595. Extruded, pultruded, molded or spiral wound derrick components shall not be painted.

* 3.1.1.2 Markings and data plates. As specified by the procuring activity for the appropriate military service (see 6.2), identification markings and data plates shall be in accordance with MIL-STD-1223. When specified (see 6.2), concealed markings shall be furnished.

3.1.1.2.1 Caution plates. Caution plates shall be furnished and mounted in conspicuous locations near the lower and upper controls and in the driver's compartment. Caution plates shall conform to composition A (class 1 or 2) or composition C of MIL-P-514. At least the following information shall appear on the plates:

(a) DERRICK MUST BE IN STOWED POSITION WHILE VEHICLE IS BEING MOVED.
(b) RATED AERIAL BASKET LOAD, HYDRAULIC OUTRIGGERS DOWN, NOT IN EXCESS OF _____ POUNDS.
(c) INSPECT AERIAL BASKET LEVELING SYSTEM.
(d) MAXIMUM VOLTAGE DERRICK AND BASKET(S) WILL WITHSTAND _____ VOLTS AC.
(e) OVERALL TRAVEL HEIGHT WHEN DERRICK IS IN STOWED POSITION.
(f) BASKET LINERS MUST BE INSTALLED WHEN OPERATING NEAR ELECTRICAL LINES.
(g) DO NOT PERMIT DIRECT CONTACT OF BASKET(S) TO GROUND WHEN OPERATING NEAR ELECTRICAL LINES.
(h) MAXIMUM SIDE SLOPES, _____ DEGREES. MAXIMUM ASCENDING SLOPES, _____ DEGREES. MAXIMUM DESCENDING SLOPES, _____ DEGREES.
(i) DO NOT DRILL HOLES IN BOOM OR BASKET TO DRAIN WATER, DUMP OUT WATER.
(j) BOOMS AND BASKET(S) MUST BE KEPT CLEAN AND FREE OF DAMPNESS.

3.1.1.2.2 Instruction plates. Plates showing schematics of all valves, valve controls, equipment, body lubrication, and complete operating instructions for operating the vehicle aerial equipment shall be fastened on the inside of one of the curbside vertical cabinet doors. Each control lever shall be marked with a plate depicting the function of the control. Individual markings shall be adjacent to each control. Plates and markings shall conform to composition A (class 1 or 2) or composition C of MIL-P-514.

3.1.1.2.3 Warning signs. Warning signs or labels, as recommended by the Manufacturers of Aerial Devices and Digger Derricks Council, shall be furnished and attached.

* 3.1.1.3 Rustproofing. When specified (see 6.2), the vehicle shall be rustproofed in accordance with FED-STD-297. When specified (see 6.2), tropical rustproofing in accordance with FED-STD-297 shall be furnished.

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3.1.1.4 Drain plugs. Drain plugs installed in the manual transmission, transfer case, and axle(s) for draining lubricants shall be of the permanent magnet type.

* 3.1.1.5 Towing devices. Not less than two hooks, loops, or pins, or a single center mounted towing eye for towing the vehicle, shall be furnished on the front of the vehicle. All towing devices shall be frame rail mounted or reinforced back to each frame rail.

* 3.1.1.6 Trailer towing package. A trailer towing package consisting of a pintle, safety chain attachment devices, a lighting receptacle, and associated reinforcements and wiring shall be installed on the rear of the vehicle. The pintle shall be of the rotating type conforming to MS51118-1. The pintle shall be installed on the chassis frame with reinforcements to transfer a vertical tongue load of not less than 4,000 pounds and a horizontal drawbar load of not less than 40,000 pounds directly to the chassis rails. The rearmost portion of the pintle shall be forward, but not more than four inches forward, of the rearmost part of the vehicle. Two trailer safety chain attachment devices, one adjacent to each side of the pintle, shall be provided. Each attachment device shall provide an ultimate strength at least equal to the GVW of the truck furnished. The attachment devices shall be capable of accommodating a standard grab hook (4-9/16 inches wide, 1-3/16 inches thick, 25/32 inch throat width) for a 5/8-inch chain. The lighting receptacle, conforming to SAE J560 with its conductors connected and color-coded or number-coded as specified therein, shall be mounted in a readily accessible location near the pintle.

3.1.1.7 Trailer lighting cable. A trailer lighting cable, conforming to SAE J1067, shall be furnished. The cable shall be coiled and shall have an extended length of not less than 110 inches. Both ends of the cable shall be equipped with a seven-circuit round plug conforming to SAE J560. The plugs shall be equipped with a grip for withdrawing from the connector sockets. The cable shall be packaged and stowed in the vehicle tool compartment.

3.1.1.8 Electric trailer brake controller. When specified (see 6.2), the vehicle shall be provided with an electric trailer brake controller. The controller shall be installed in the truck cab and wired through the SAE J560 lighting receptacle. The controller shall operate on the 12-volt electrical system of the vehicle and shall include load control, hand lever and accessories.

3.1.1.9 Spark arrester. As specified herein (see 3.4.4), the vehicle shall be furnished with a spark arrester.

* 3.1.1.10 Wheel splash and stone throw protection. The vehicle shall have rigid splash shields ahead of the rear wheels and rubber mud flaps to the rear of the rear wheels. All tilt cabs shall have rubber mud flaps to the rear of the front wheels. A metal strip not less than 0.1250 inch thick and not less than one inch wide, extending the entire width of the mud flap,

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shall be installed to prevent the bolt heads or bolt nuts from damaging the mud flap. As an alternate method of attaching the mud flaps, tabs or clips with minimum surface contact dimensions of 1 inch high by 1.25 inch wide by 0.09375 inch thick shall be furnished at each bolt. All splash shield and mud flap installations, front and rear, shall conform to the rear wheel splash and stone throw protection provisions of SAE J682.

3.1.1.11 Brushguards. Model 4x4 and 6x6 shall be equipped with a radiator and headlamp brushguard. When the headlamps are recessed into and protected by the front bumper, a headlamp brushguard is not required.

* 3.1.1.12 Brake lights. At least one pair of brake lights shall override the four-way emergency flasher. Modifications to the manufacturer's standard product to accommodate this requirement shall not compromise conformance to any Federal Motor Carrier Safety Regulation referenced herein or to any Federal Motor Vehicle Safety Standard. If additional lights are added to the vehicle, the lights shall be selected from the chassis manufacturer's standard matching hardware.

* 3.1.1.13 4x4 or 6x6 conversion. The chassis manufacturer's standard 4x2 or 6x4 truck chassis may be modified to provide all wheel drive conforming to the requirements specified herein if:

(a) The conversion axle manufacturer's engineering department specifically approves and certifies that all such modifications meet the design requirements and standards of the conversion axle manufacturer. Certification shall be based on both design analysis and proving ground test reports, which shall be made available to the engineering and quality assurance offices of the procuring activity.
(b) The chassis manufacturer's front axle before conversion has the same load rating as the conversion axle to be installed.
(c) Components used in the all wheel drive conversion are of current production.
(d) Components used in the all wheel drive conversion are approved for the conversion application by the component manufacturers.
(e) The converted vehicle is certified to conform to Federal Motor Vehicle Safety Standard No. 121, by the intermediate or final manufacturer.
(f) Replacement headlights, if required to be added to meet the height requirements of Federal Motor Vehicle Safety Standard No. 108, are equivalent in mounting, protection, and range and precision of adjustment to the chassis manufacturer's original standard headlights.
(g) Unused headlight cavities are covered in a neat workmanlike manner, treated and painted to match the chassis cab color with treatment and painting equivalent to the chassis manufacturer's process for the remainder of the chassis cab. Cavities and their covers shall be rustproofed in accordance with 3.1.1.3.

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(h) Complete installation drawings are available to the procuring activity.

3.2 General design.

3.2.1 Federal Motor Vehicle Safety Standards. The vehicle and furnished accessories shall comply with all Federal Motor Vehicle Safety Standards in effect on the date of manufacture.

3.2.2 Air pollution control. The vehicle shall comply with the EPA Regulations governing Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines in effect on the date of manufacture. In addition, vehicles destined for California shall comply with the State of California regulations governing air pollution control in effect on the date of manufacture.

* 3.2.3 Sound level. The interior sound level shall not exceed 84db(A). The vehicle exterior drive-by sound level shall conform to EPA Noise Emission Standards for Transportation Equipment - Medium and Heavy Trucks.

3.2.4 Safety. The vehicle and rotating derrick shall comply with ANSI A92.2 and all OSHA requirements, including OSHA 1926.556.

3.2.5 Curb weight. The curb weight shall include the weight of the chassis and cab, with all attachments, accessories, equipment (but excluding tools and augers), body, derrick and a full complement of fuel, lubricants, and coolant.

3.2.6 Gross vehicle weight. The gross vehicle weight (GVW) shall consist of curb weight, operator weight (computed at 175 pounds), and a payload to provide not less than the specified GVW; or a payload of not less than 2,500 pounds, whichever provides the greater GVW.

3.2.6.1 Weight distribution. The distribution of the GVW for the purpose of establishing suspension, axle, and tire capacities shall be determined with the payload uniformly distributed over the vehicle body load area. In addition, front suspension, axle, and tire capacities shall be not less than 9,000 pounds, measured at the ground.

3.2.7 Ratings. The vehicle and component ratings shall be the manufacturer's published ratings. Vehicle and component ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verification of ratings shall be available to the engineering office of the procuring activity. Minimum GVW ratings shall conform to 1.2.1 for the specified class of vehicle. In addition, the actual GVW rating of the vehicle furnished shall exceed the minimum GVW specified when required to provide a payload of not less than 2,500 pounds evenly distributed over the load area.

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* 3.2.8 Overall width. The overall width of the vehicle, exclusive of tires and safety related items such as mirrors, lights and reflectors shall be not more than 96 inches. The width over the tires shall be not more than 100 inches.

3.2.9 Accessibility. The design of the vehicle and optional equipment shall permit access for routine servicing and shall permit access for replacement and adjustment of component parts and accessories with minimal disturbance of other components and systems.

* 3.2.10 Asbestos. Except as permitted in 3.4.11, asbestos materials shall not be used in any form in any part of the vehicle.

3.3 Performance.

3.3.1 Speeds. High and low speed requirements shall be met with the vehicle loaded to required GVW. For model 4x4 and model 6x6 vehicles, performance requirements shall be met with the front wheel drive disengaged; except that when equipped with interaxle compensating devices, the vehicle shall meet the performance requirements with the front wheel drive engaged.

* 3.3.1.1 High speed gradeability. The vehicle shall ascend a 1.0 percent grade at 50 miles per hour (mph). Gradeability requirements shall be met with the main transmission in direct drive, and when a multispeed axle or auxiliary transmission is furnished, with the axle or auxiliary transmission in high speed range. Gradeability shall be verified with calculations in accordance with SAE J688 (see 6.3).

3.3.1.2 Low speed. Low speed for vehicles with a manual transmission shall be calculated with the engine operating at not less than 35 percent of the recommended governed speed, and shall provide a vehicle speed of not more than 3 mph.

* 3.3.1.3 Maximum geared speed. The maximum geared speed at engine governed speed shall be not less than 52 mph for model 4x4 and 6x6 vehicles and not less than 58 mph for model 4x2 and 6x4 vehicles. Conformance to geared speed specified shall be determined by calculating in accordance with the following formula:

$$\text{Maximum geared speed (mph)} = \frac{\text{Governed speed (rpm)}}{\text{Total gear reduction x tire factor (see 6.3).}}$$

3.3.2 Service brakes. The service brakes shall control and hold the vehicle on a 30 percent grade when loaded to its specified GVW. The service brakes shall stop the vehicle, loaded to specified GVW, within the stopping distance requirements of Federal Motor Carrier Safety Regulation 393.52.

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3.3.3 Height and reach. The derrick shall provide for hydraulically powered movement from horizontal to not less than 75 degrees above horizontal and return, continuous rotation about a vertical axis in both directions, and telescopic extension and retraction. For each telescoping section position listed in table III at not more than the corresponding angle specified in table III, the derrick height, from the ground to the centerline of the upper head sheave, shall be not less than that specified in table III. The basket floor to ground height shall be not less than specified in table III.

3.3.4 Vertical loads. For each telescoping section position listed in table III, the derrick shall be capable of suspending, on the winch line over the upper head sheave, the corresponding load at not more than the specified angle and at all greater angles attainable. The derrick shall be capable of elevating from and lowering back to the angle specified with the load but shall be incapable of elevating from a horizontal position with loads that would result in permanent deformation. The derrick shall be capable of retracting (from extended positions) and returning back to extended position with the load. All operations of rotating, elevating or lowering, and extending or retracting shall be possible singly and in all combinations, all while both loaded and unloaded. Operations as specified above shall be possible at all rotational positions of the derrick without any vehicle wheel leaving the ground and without the permanent deformation of any part.

TABLE III. Height, reach and loads.

Telescoping section position	Degrees from horizontal	Height from ground (feet) size		Load (pounds) size	
		55	70	55	70
Without fiberglass extension:					
1. Retracted	75	22	26	9,000	12,000
2. Extended	75	30	35	6,500	8,000
3. Extended	45	24.5	28	3,400	3,400
4. Horizontal extended	0	10	10	1,800	2,200
With derrick and fiberglass extension extended:					
1. Without baskets	75	36.5	44.5	4,500	6,300
	45	29.5	35	2,000	2,500
	0	10	10	1,350	1,700
2. Basket floor to ground height	75	32	40	300	300
	0	-	-	300	300

3.3.5 Rotation. The turret assembly shall rotate the loads specified in 3.3.4 at a speed of not less than 0.5 revolution per minute (rpm) and not more than 1.5 rpm with the engine pushbutton throttle advanced (see 3.12). The turret assembly shall be capable of smoothly

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rotating the derrick with the vehicle up and down a 15 percent grade at all derrick positions and loads listed in table III without any vehicle wheel leaving the ground and without permanent deformation of any part.

3.3.6 Horizontal load. The turret and derrick shall be capable of rotating against a horizontal load, at right angles to the derrick, of sufficient force to produce a torque of not less than 21,000 foot-pounds for size 55 and 37,000 foot-pounds for size 70 without opening the turret rotation hydraulic relief valves and without permanent deformation of any part. The turret shall be provided with hydraulic relief valves preventing rotation against a horizontal load, at right angles to the derrick, that would result in permanent deformation of any part.

3.3.7 Position holding ability. The turret and derrick assembly shall be capable of remaining in a fixed position and of controlled lowering when the hydraulic motor is de-energized. With the vehicle longitudinal on a 15 percent grade and the derrick transverse, the turret and derrick shall hold, in all positions and load combinations specified in 3.3.4, without more than 1 inch per hour horizontal creep and without more than 2 inches per hour vertical settling, both measured at the centerline of the upper head sheave.

3.3.8 Performance with aerial basket(s). With the basket(s) loaded with twice the payload specified in 3.7.3, the derrick shall perform as specified in 3.3.3 through 3.3.7 through all positions it is capable of attaining without any wheel of the vehicle leaving the ground and without the permanent deformation of any part.

3.4 Chassis components.

3.4.1 Engine. The engine furnished shall be the chassis manufacturer's standard or optional engine for the commercial model vehicle which meets or exceeds the requirements of this specification.

* 3.4.1.1 Diesel engine. Unless otherwise specified (see 3.4.1.2), the vehicle shall be equipped with a liquid cooled, compression ignition, two stroke or four stroke cycle diesel engine, with not less than four cylinders. The engine net horsepower used in performance prediction calculations shall be determined in accordance with SAE J1349. The engine shall demonstrate the performance characteristics specified herein when using diesel fuel conforming to VV-F-800. In addition, when specified (see 6.2), the engine shall operate satisfactorily on grade JP-4 and grade JP-5 fuel conforming to MIL-T-5624 under emergency, short duration conditions and on grade JP-8 fuel conforming to MIL-T-83133 under normal conditions. A power loss when operating on JP-4, JP-5 or JP-8 is acceptable. A test report shall be available showing that a representative engine of the same model has been tested and certified to meet the requirements of NATO AEP-5 (400 hours dynamometer test to assure JP-8 compatibility.)

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3.4.1.2 Gasoline engine. When specified (see 6.2), for class A only, the vehicle shall be equipped with a liquid cooled, internal combustion, four-stroke cycle, gasoline engine with not less than six cylinders. The engine furnished shall produce the required vehicle performance when operated on unleaded fuel with a research octane rating of 91 at engine speed not more than the manufacturer's recommended operating speed. The engine shall be capable of warranted operation on unleaded fuel when used in accordance with the operator's manual. Net horsepower figures used in performance prediction calculations shall be determined in accordance with SAE J1349. Unless otherwise specified or required by foreign laws, vehicles for overseas use shall be capable of accepting and operating on leaded gasoline.

3.4.1.3 Oil filter. A full flow type oil filter shall be furnished.

3.4.1.4 Governor. An engine governor shall be furnished and set and sealed to limit the engine to the engine manufacturer's maximum recommended operating speed.

* 3.4.1.5 Cooling system. The cooling system shall maintain the engine coolant at a temperature below the boiling point with the vehicle loaded to rated GVW and operated at an altitude of 10,000 feet above sea level or in an ambient air temperature of not less than 125 degrees Fahrenheit (ºF). The coolant system shall include a surge tank or a coolant recovery reservoir of not less than two-quart capacity. On tilt cab models, a radiator servicing access door shall be provided to allow verification of the coolant level.

* 3.4.1.6 Coolant temperature control. Thermostatic control of the engine coolant temperature shall be provided. On diesel engine driven vehicles, the control shall include complete thermostatic control of all coolant flow through the radiator.

* 3.4.1.7 Fan clutch. A fan clutch shall be provided. The fan clutch shall reduce the fan speed automatically when the fan is not required for engine cooling. The fan clutch shall be asbestos free.

3.4.1.8 Silicone rubber hoses. When specified (see 6.2), silicone rubber radiator and heater hoses shall be furnished.

* 3.4.1.9 Power plant heaters and fuel warmer. When specified (see 6.2), a coolant heater, an engine oil heater, and a fuel warmer (for diesel engine driven vehicles only) shall be provided. Unless otherwise specified (see 6.2), a battery heater shall be provided when power plant heaters are specified. Heaters shall operate on 110-volt alternating current (ac), and shall be wired through a junction block, including a fuse or circuit breaker, to a single three-pronged (male), weatherproof, slave receptacle for receiving external power and grounding the vehicle. A three-wire connecting cable, 25 feet long and of adequate line capacity to supply power for all heater units simultaneously, shall be furnished. The connecting cable shall include a matching

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female connector at the vehicle end and a standard, weatherproof, three-pronged (two power plus one ground) male connector at the other end. Electrical apparatus shall conform to Federal Motor Carrier Safety Regulation 393.77(c)(7). Electrical insulation of the connecting cable shall withstand normal operating stresses in low ambient air temperatures (down to minus 60ºF) without cracking or loss of dielectric capacity. All heater lead wires shall be installed without interfering with vehicle component operation and without loose excess wire. Provision for stowing the cable shall be provided inside the vehicle cab. Heaters shall be furnished as follows:

(a) A coolant heater, 1500-watt minimum rating, shall be installed in the engine block or lower coolant inlet hose. Engine coolant thermostat with an operating range of 70°F to 195°F shall be installed
(b) An immersion type engine oil heater, 300-watt minimum rating, with 170ºF to 195°F thermostat, shall be installed in the oil pan through any convenient opening
(c) The battery heater shall have a capacity adequate to maintain the battery electrolyte at a temperature of not less than 10ºF during vehicle exposure in ambient air temperatures as low as minus 60ºF, and shall embody a thermostat to limit the temperature of the electrolyte to not more than 80°F
(d) When an auxiliary gasoline engine (see 3.15) is provided, a 300-watt immersion type heater with 170ºF to 195ºF thermostat shall be installed in the oil pan through any convenient opening
(e) A fuel warmer or preheater shall be provided on diesel engine driven vehicles to prevent clogging of fuel filters due to wax crystallization in the fuel. The fuel warmer shall use engine coolant to transfer sufficient heat to the diesel fuel to heat it from an inlet temperature of minus 40ºF to an outlet temperature of plus 9°F, with a flow rate not less than the maximum fuel demand of the engine fuel system. A coolant shutoff valve shall be provided for the coolant inlet side of the fuel warmer unit.

* 3.4.1.10 Fuel fired engine preheater. When specified (see 6.2), for diesel engine driven vehicles, a diesel fuel fired engine water heater shall be furnished to preheat the engine. The heater shall include a timer, a thermostat and a circulating pump and shall be connected to the engine coolant system. The heater shall be capable of starting and operating at minus 60ºF and shall heat the engine to plus 40ºF from minus 60ºF in not more than 1 hour. The system shall be equipped with a light, visible to the driver, to indicate that the preheater is operating.

3.4.2 Electrical system. The electrical system shall be in accordance with Federal Motor Carrier Safety Regulations 393.27 through 393.31 and 393.33.

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* 3.4.2.1 Starting system (diesel engines). For diesel engine driven vehicles, a 12- or 24-volt direct current (dc) starting system, with a 12-volt dc lighting system, and not less than a 100-ampere alternator, shall be furnished. Engine starting equipment shall include an ether starting system, glow plug or electric grid heater. If an ether system is furnished in lieu of a glow plug or electric grid heater, it shall be of the measured shot type. The measured shot type ether system shall be key operated or manually operated from the driver's compartment and shall be inoperative with the engine warm. Complete provisions for a replaceable ether reservoir of not less than 12 fluid ounces shall be furnished. A reservoir need not be furnished.

3.4.2.2 Ignition system (gasoline engines). For gasoline engine driven vehicles, a 12-volt dc ignition system shall be furnished. An alternator of not less than 60-ampere rated capacity and which provides not less than 16 amperes of dc output at normal engine idle speed shall be furnished.

* 3.4.2.3 Lighting. All vehicle lights, reflectors, and wiring shall conform to Federal Motor Carrier Safety Regulations 393.12, 393.19, 393.20 and 393.22 through 393.26(d). Lights and reflectors shall not be mounted on vertical surface of rub rails (unless recessed and fully protected) or on vehicle bumpers.

* 3.4.2.4 Turn signals. Turn signal lamps shall conform to SAE J588. Operating units shall conform to SAE J589 and shall be mounted on the steering column. Turn signals shall be installed in accordance with SAE J588. Turn signal operating units shall have a visible and audible type flash indicator.

3.4.2.5 Batteries. Each battery shall be of 12-volt potential. The total reserve capacity rating and the total cold cranking rating at 0ºF, both measured in accordance with SAE J537, shall be not less than specified in table IV. The batteries shall be of the maintenance-free type, having the maintenance-free characteristics listed in W-B-131.

TABLE IV. Batteries.

Engine type	Reserve capacity (minutes)	Cold cranking (amperes)
Diesel (under 220 gross horsepower)	320	1,200
Diesel (220 gross horsepower or over)	480	1,740
Gasoline	100	450

3.4.2.6 Radio interference suppression. The vehicle shall be suppressed to limit electromagnetic radiation in accordance with SAE J551. Any body equipment emitting electromagnetic radiation shall be suppressed to the same level as the vehicle chassis.

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3.4.3 Fuel system. The fuel system shall conform to Federal Motor Carrier Safety Regulations 393.65 and 393.67.

3.4.3.1 Air cleaner. An air cleaner shall be furnished.

3.4.3.2 Fuel tank(s). Fuel tank(s) shall have not less than 100 gallons total capacity (85 gallons for vehicles destined for California). When more than one tank is furnished on diesel engine driven vehicles, means shall be provided to assure an equalized fuel level in both tanks. When more than one tank is furnished on gasoline engine driven vehicles, a selector valve connecting either tank to the engine fuel intake shall be provided and a means shall be provided to monitor the fuel level of either tank from a single fuel gage.

* 3.4.3.3 Fuel and water separator. The manufacturer's standard or optional fuel filter shall be provided. When specified (see 6.2), a fuel and water separator shall be furnished. The separator shall include a water coalescer and a drain valve. When a separator is required, a combination filter/separator unit may be furnished.

3.4.4 Exhaust system. The exhaust system shall conform to Federal Motor Carrier Safety Regulation 393.83. A vertical exhaust system shall not be furnished. The exhaust outlet shall be located to direct the exhaust gases away from the control console, in all positions of rotation. A spark arrester shall be furnished on all vehicles except on turbocharged engines. The spark arrester shall have an 80 percent arresting efficiency when rated in accordance with SAE J350.

3.4.4.1 Exhaust system safety shield. The exhaust system or hydraulic system shall be shielded so that, in the event of a hydraulic line rupture or hydraulic fluid leak, the hydraulic fluid will not come into contact with any part of the vehicle exhaust system.

3.4.5 Gear train.

3.4.5.1 Manual transmission. Unless otherwise specified (see 3.4.5.5) a manually shifted transmission shall be provided. The input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine. The transmission shall provide for maximum ease of shifting in all speeds. The transmission shall be provided with two power takeoff openings conforming to SAE J704. One power takeoff opening shall be of the heavy duty type.

3.4.5.2 Speeds. The vehicle with manual transmission shall be provided with not less than the number of forward speeds specified in table V. A two speed axle, a multispeed transmission or an auxiliary transmission shall be furnished. The gear ratios in the transmission and other gear selection equipment shall be matched to provide a progressive shifting pattern throughout the complete range and shall provide the required vehicle performance specified in 3.3.1 through 3.3.1.3.

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TABLE V. Vehicle forward speeds.

Vehicle model	No. of forward speeds
4x2	8
4x4	9
6x4	9
6x6	9

* 3.4.5.3 Transfer case (models 4x4 and 6x6). Model 4x4 and 6x6 shall be provided with a transfer case. Unless the transfer case is equipped with devices which compensate for differential torque and speeds between the front and rear axles, the transfer case shall provide for driver selection of either two-wheel, four-wheel, or six-wheel drive, as applicable. When interaxle compensating devices are furnished, they shall provide for the positive transfer of power to all driving axles. When a two-speed transfer case is furnished, the speedometer shall read accurate speed (mph) with the speed selector in high or low range.

3.4.5.4 Clutch. The clutch shall be largest capacity clutch offered for the type, class, and model vehicle and engine furnished, with the clutch torque capacity exceeding the maximum delivered engine torque.

* 3.4.5.5 Automatic or semiautomatic transmission. When specified (see 6.2), the vehicle shall be provided with an automatic or semiautomatic transmission. The transmission shall include a hydraulic torque converter and not less than four forward gear ratios for class A and five forward gear ratios for class B and C. Normal driving range selector position shall provide not less than four gear ratios without movement of the selector. The transmission shall be provided with an SAE J704 power takeoff opening. The net torque capacity and the net power rating of the transmission shall exceed the output ratings of the engine.

3.4.5.6 Power takeoff. A power takeoff shall be furnished, and shall be of a rated capacity to operate powered equipment. Controls to operate the power takeoff shall be located in the truck cab accessible to the seated driver. A caution plate or decal reading “DO NOT OPERATE VEHICLE AT HIGHWAY SPEEDS WITH POWER TAKEOFF ENGAGED” shall be installed so as to be readily visible to the seated driver.

3.4.6 Driveline components. The driveline components shall be adequate to transmit the maximum delivered torque of the engine, as developed through the maximum gear train reduction, in proportion to the number of driving axles furnished.

3.4.7 Frame. The chassis frame shall be the manufacturer’s standard for the type and class of vehicle furnished. The vehicle shall have a frame with the reinforcement extending at least from the rear of the front suspension rear hanger bracket to the front of the rear spring front hanger bracket on model 4x2 and model 4x4 vehicles, and to the bogie trunnion mounting

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bracket on model 6x4 and model 6x6 vehicles. Reinforcement shall provide sufficient structural strength in the chassis frame to adequately distribute and support the additional loads imposed by the rotating hydraulic derrick and the outrigger supports. The reinforcement shall provide sufficient structural strength to the chassis frame through increased resisting bending moment to at least equal the loads imposed on the vehicle body loaded to provide specified GVW. The chassis frame rails shall not project beyond the rear end of the body. The chassis frame for type I and type III, size 70 vehicles shall extend to under the rearmost outrigger. The chassis frame for type II vehicles shall extend to under the rear corner mounted derrick.

3.4.8 Suspension. The vehicle shall be equipped with a suspension system with components having a rated capacity at least equal to the load imposed on each member, measured at the ground, with the vehicle loaded to the required GVW. In addition, the rated capacity of the front suspension shall be not less than as specified in 3.2.6.1. When the suspension capacity is rated at the spring pads, unsprung weight shall be deducted. The vehicle shall be equipped with hydraulic, double-acting shock absorbers at the front wheels. Suspension system ratings shall be adequate for the loads imposed during all specified operating conditions.

3.4.9 Axles. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground, with the vehicle loaded to the required GVW. In addition, the front axle rating shall be at least equal to that specified in 3.2.6.1. Gear ratio(s) shall provide the performance specified in 3.3.1 through 3.3.1.3. When specified (see 6.2), the wheel bearings and axle spindles shall be oil lubricated. Except for driving axles, the hubcaps shall have a window for visual determination of oil level. Provisions for venting or withstanding internal pressure buildup and for replenishing the oil supply shall be provided.

3.4.9.1 Two-speed axle. When specified (see 6.2), or when required to conform to 3.4.5.2, models 4x2 and 6x4 shall be furnished with a two-speed axle. The two-speed axle shall be equipped with electric, vacuum, or air shift, and provided with ratios which shall permit proper gear splitting. The gear ratios shall provide the performance specified in 3.3.1 through 3.3.1.3.

3.4.9.2 Rear bogie (models 6x4 and 6x6). The rear bogie shall be of the four-wheel type, complete with axles, springs, torque rods, and all other necessary parts. The bogie shall be provided with a means of permitting differential action between the two axles and a manually or automatically controlled lockout assuring equal power to each rear axle. The manual lockout control used shall be located in the truck cab. Axle gear ratios shall provide the performance specified in 3.3.1 through 3.3.1.3.

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3.4.9.3 Traction control. When specified (see 6.2), the vehicle shall be furnished with a traction control on the rear axle for models 4x2 and 4x4 and on either rear axle on models 6x4 and 6x6. The traction control shall actuate automatically to ensure that power is transmitted to the wheels having traction when the opposite wheel loses traction. Maximum traction capabilities shall be maintained at all times under each drive wheel for the life of the vehicle.

* 3.4.10 Wheels, rims tires and tubes. Unless wide base tires are specified, the vehicle shall be equipped with single front and dual rear wheels. Rims and tire ratings shall conform to Tire and Rim Association or European Tyre and Rim Technical Organisation recommendations for the type and size of tires furnished. Except when tube type tires are specified in acquisition documents, multipiece rims shall not be furnished on front axles rated at 14,000 pounds or less, on single rear axles rated at 23,000 pounds or less, or on tandem axles. Rim and disc wheel sizes shall be the same for all wheels on each vehicle. Tire size and load range (ply rating) shall be the same for all tires on each vehicle. When specified (see 6.2), wide base type tires and wheels for the front and rear axles shall be provided in lieu of conventional front and dual rear wheels and tires. Wide base wheels shall be interchangeable without the use of an adapter. When specified (see 6.2), 10-stud disc type wheels shall be furnished.

* 3.4.10.1 Tires. Steel belted radial or, when specified (see 6.2), bias ply tires shall be furnished. Tires shall have highway tread for model 4x2 and model 6x4 vehicles, and nondirectional mud and snow tread for model 4x4 and model 6x6 vehicles. Tires shall be of the tubeless type or, for axles where multipiece rims are permitted (see 3.4.10), may be of the tube type. When tube type tires are permitted herein or specified, multipiece rims may be furnished. Tires shall be of rated capacity at least equal to the load imposed on each tire, measured at each wheel, at the ground, with the vehicle loaded to specified GVW. In addition, front tires shall be of a rated capacity at least equal to that specified in 3.2.6.1. Tires and tire size designation systems shall conform to the Tire and Rim Association or to The European Tyre and Rim Technical Organization recommendations.

3.4.10.2 Inner tubes. When tube type tires are permitted herein or specified in acquisition documents, inner tubes shall be of the heavy duty type, and shall be of proper size for the tires furnished. Tire flaps shall be provided for tube type tires in accordance with The Tire and Rim Association recommendations.

3.4.10.3 Carrier for spare tire assembly. When specified (see 6.2), a carrier for a spare wheel or rim and tire assembly shall be installed in a readily accessible location on the vehicle. Threaded fasteners, when used to secure the spare tire in the carrier, shall be constructed of or plated with corrosion-resistant material. The carrier design shall enable removal and mounting of a spare assembly using only the tools specified in 3.4.16.1.

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3.4.10.4 Spare wheel or rim. When specified (see 6.2), a spare wheel or rim shall be provided. When a spare tire assembly is specified (see 3.4.10.5), a spare wheel or rim shall be furnished.

3.4.10.5 Spare tire assembly. When specified (see 6.2), a spare tire assembly shall be furnished. The spare tire assembly shall include an inflated spare tire mounted on the spare wheel or rim. The spare tire shall be of the same size, tread design and load range (ply rating) as the tires furnished on the vehicle.

3.4.10.6 Tire chain clearance. Tire chain clearance in accordance with SAE J683 shall be provided. Allowance for spring deflection shall be included.

* 3.4.11 Brakes. The brakes shall conform to Federal Motor Carrier Safety Regulations 393.40 through 393.42(b), 393.43, and 393.46 through 393.52. Brake linings shall be of nonasbestos material on single axles rated at 23,000 pounds or less and on tandem axles.

3.4.11.1 Service brakes. The vehicle shall be equipped with full-air brakes on all wheels. The braking system, complete with all necessary components, shall include:

(a) Air compressor, unloader-head type, engine driven and engine lubricated, air or water cooled, and having a capacity of not less than 7.25 cubic feet per minute (cfm) for model 4x2 and model 4x4 vehicles and 12 cfm for model 6x4 and 6x6 vehicles
(b) Air storage reservoir(s), each tank equipped with drain, and with safety and check valves between the compressor and the last reservoir tank
(c) Foot control, suspended or treadle type
(d) Air control valves
(e) Air pressure gage visible to the driver
(f) Low air pressure warning, visible and audible
(g) Service brake stop lamp switch
(h) Automatic moisture ejector
* (i) For model 4x2 vehicles with rear axles rated at 23,000 pounds or less, and for 6x4 vehicles, automatic slack adjusters on cam type brakes or internal self-adjusting brakes on wedge and disc type brakes on all axles. Vehicles without this feature on one axle (rear axle rated at over 23,000 pounds) shall not have it on any axle.

3.4.11.1.1 Air dryer. A replaceable cartridge desiccant type air dryer shall be installed in the air brake system. The dryer shall have the capability of removing not less than 95 percent of the moisture in the air being dried. The dryer shall have a pre-cooler and a filter to screen out oil and solid contaminants. The dryer shall have an automatic self-cleaning cycle and a thermostatically controlled heater to prevent icing of the purge valve.

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3.4.11.2 Trailer brake control system. When specified (see 6.2), a trailer brake control system shall be furnished in addition to the components specified in 3.4.11.1 and 3.4.11.1.1. The trailer brake control system shall include:

(a) Identification of emergency and service lines
(b) Coincident control of trailer brakes with the prime mover foot control
(c) Independent hand control for the trailer brakes
(d) Prime mover protection valve with a dash control and automatic break-away feature
(e) Trailer stoplight control operable with a foot brake and with hand control for the trailer brakes
(f) Air connectors for trailers with SAE J318 gladhand couplers mounted at the rear of the vehicle, located to prevent interference with trailer
(g) Two SAE J844 coiled air hoses, each not less than 110 inches long when fully extended, with SAE J318 gladhand couplers on both ends of hoses. The hoses shall be packaged and stowed in the vehicle tool compartment for shipment
(h) Dummy gladhand couplers with chains
(i) Prime mover only parking brake valve to permit prime mover parking brakes to be applied while charging the trailer air brake system.

3.4.11.3 Brake controls for use from a towing vehicle. When specified (see 6.2), the vehicle shall be furnished with a system for controlling the brakes from a towing vehicle (wrecker). The installation shall be complete with air brake couplers, a relay emergency valve with no-bleed-back feature (except when a spring applied emergency brake is furnished), additional air lines and fittings. The service and emergency couplers shall be mounted on the front in a protected position providing for ready attachment of air hoses from a towing vehicle. The service and emergency couplers shall be identified and provided with dummy gladhand couplers with chains. The system shall not compromise conformance to any Federal Motor Carrier Safety Regulation referenced herein or to any Federal Motor Vehicle Safety Standard.

* 3.4.12 Cab. Unless otherwise specified, the chassis manufacturer shall furnish any type of his standard or optional full width cab. When specified (see 6.2), a cab with a forward tilting hood and fender assembly, including tilting and locking mechanism, shall be furnished. When specified (see 6.2), type II and type III vehicles shall be furnished with a tilt type cab having a tilting and locking mechanism. Tilt cab features shall be in accordance with the Employee Safety and Health Standards of Federal Motor Carrier Safety Regulation 399, including the first step height, which shall be not more than 24 inches. The cab doors shall be equipped with locks operable from inside the cab through mechanical linkage and with at least the curbside door equipped with an external, key operated lock. Drip rails shall be installed above the cab doors. Safety grips shall be provided on each side of the cab for assisting personnel in entering and

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leaving the cab. When the front tires extend beyond the cab fenders, rubber fender extensions extending at least to the outside of the tire tread shall be furnished. When tilt type cabs are furnished, provisions to facilitate cleaning the windshield shall be provided by means of a bumper step or bumper step cutouts and a grab handle located under the windshield. When step height exceeds 24 inches, a secondary step shall be provided.

3.4.12.1 Cab interior. Unless otherwise specified, the cab shall have an upholstered, full width, adjustable seat and back or individual, adjustable, driver's seat and individual passenger seat. When specified (see 6.2), an individual, adjustable, driver's seat and an individual passenger seat shall be provided. When specified (see 6.2), the driver's seat shall be the manufacturer's standard air ride suspension type. The color of the upholstery and the interior finish shall be compatible with the exterior color (see 3.1.1.1). White upholstery shall not be furnished. Interior lighting shall be provided. Two pairs of seat belts shall be installed.

* 3.4.12.2 Crew cab. When specified (see 6.2), a four-door, full width crew cab shall be furnished in lieu of the standard full width cab. The cab shall be equipped with two upholstered, full width seats and backs. The front seat shall be adjustable. Three pairs of seat belts shall be installed for both the front and rear seats. Cab doors shall be equipped with locks operable from inside the cab through mechanical linkage, with both front doors equipped with an external key operated lock. Cab doors shall have windows with crank operated window regulators. A rear window shall be provided. Interior lighting shall be provided. Safety grips or grab handles shall be provided at each door of the cab to assist personnel climbing into the cab. The cab roof shall be of one-piece construction; or, if welded, the roof shall give the appearance of one piece, with weld seams being continuous, waterproof , and free of visible seams, bumps or protrusions. Full length drip moldings shall be mounted above the doors. When a crew cab is furnished, the cab-to-axle dimension shall be measured from the back face of the crew cab.

3.4.12.3 Crew cab seat spacing. With the front seat adjusted to the extreme forward position there shall be not less than 30 inches, measured in a horizontal plane, between the front of the rear seat-back and the rear of the front seat-back. A kick-space height of not less than 2.75 inches shall be maintained between the floor and the rear of the front seat in all positions of adjustment. Leg room and space forward of the front seat shall be equivalent to that provided ahead of the seat in a two-door standard cab.

3.4.13 Steering. Power steering shall be furnished.

3.4.14 Windshield wipers and washers. The vehicle shall be equipped with dual windshield wipers and windshield washers. The windshield wipers shall be multispeed type operated by either air or electric motor(s).

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3.4.15 Bumper. Unless the bumper is an integral part of the vehicle cab, or unless a front mounted winch is specified (see 3.4.22), a channel type front bumper shall be provided.

3.4.15.1 Rear end protection. The rear end of the vehicle shall be protected in accordance with Federal Motor Carrier Safety Regulation 393.86. A rear bumper shall be provided (see 3.5.1).

3.4.16 Tool stowage. Stowage space of sufficient size to accommodate a vehicle jack, hand tools, antiskid chains, and emergency reflective triangles shall be furnished for retaining equipment during vehicle operation. Stowage space for these tools may be furnished inside the cab or body cabinets. When stowage space for these tools is located outside of the cab or body, it shall be weatherproof and shall provide for locking with a padlock.

3.4.16.1 Tools. When specified (see 6.2), each vehicle shall be furnished with tools required for exchanging a mounted tire assembly with the spare assembly, and shall include at least a hydraulic jack, jack handle, and wheel-nut wrench. The jack shall be of such closed height as to permit its location under the axle, or other satisfactory lift point, at any wheel with the tire flat. The jack, without blocking, shall be capable of raising any wheel of the loaded vehicle to a height adequate to permit removal and replacement of a wheel and tire assembly.

* 3.4.17 Heater and defroster. The vehicle shall be provided with a hot water heater with fresh air intakes and discharge outlets to the floor and to windshield defroster louvers. The heater shall be complete with blower and mounted controls convenient to the driver.

* 3.4.18 Controls and operating mechanisms. All controls and operating mechanisms shall be located for left-hand drive. Controls shall be complete and conveniently operable by the driver. Lever controls shall be designed and located to permit easy entrance and exit of the operator to and from the driver's compartment. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing. All instruments shall be visible to the driver when seated in the driving position.

3.4.19 Accessories and equipment. The chassis equipment shall be complete with all accessories furnished as standard equipment by the manufacturer. The following minimum equipment shall be furnished:

(a) Key-operated ignition switch
(b) Ammeter, charging indicator or voltmeter
(c) Fuel gage
(d) Oil pressure gage or red indicator warning light
* (e) Engine coolant temperature gage or high coolant temperature or low coolant level red indicator warning light
* (f) High coolant temperature or low coolant level alarm buzzer
(g) Speedometer with recording odometer

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(h) Tachometer
(i) Ash receptacle
(j) Dual sunvisors
(k) Armrest on driver's side
(l) Driver's compartment ventilator other than window.

* 3.4.19.1 Rearview mirrors. Two outside rearview mirrors shall be furnished, one mounted on each side of the cab. Each mirror shall be of the combination type having flat and convex areas enclosed in a common housing. The flat portion shall have not less than 50 square inches of reflective area. The convex portion shall have not less than 25 square inches of reflective area. The mirrors shall have not less than two supporting arms.

3.4.20 Horn. The manufacturer's standard electric horn shall be furnished. When specified (see 6.2), in addition, an air operated horn shall be furnished.

3.4.21 Engine hour meter. An engine hour meter having a totalizing mechanism of not less than 9,999 hours shall be furnished for the chassis engine to register accurately the number of hours of operating time. The meter shall be of rugged construction to ensure continuous trouble-free performance under severe operating conditions. The engine hour meter shall be mounted on the cab instrument panel or in the engine compartment in a readable location.

3.4.22 Front mounted winch. When specified (see 6.2), the vehicle shall be furnished with a winch assembly, mounted on the front of the vehicle and powered by the vehicle's hydraulic system (see 3.12). The winch shall conform to SAE J706. The winch shall be the single drum type for hoisting and negotiating purposes and shall have one forward and one reverse speed. Dual winch controls shall be furnished, one inside the cab and one at the front bumper. The cab winch controls shall be so located as to provide no interference with the entrance or exit of the driver. The winch shall have a rated single line pull capacity of not less than 20,000 pounds on the bare drum. The winch shall be equipped with a hydraulic relief valve set to restrict the line pull capacity to 22,000 pounds. The winch line speed on the bare drum shall be between 15 and 30 feet per minute at an engine speed equal to 35 percent of engine governed speed. The winch shall be wound with not less than 185 feet of 0.625 inch diameter, preformed, 6x37, improved plow steel, independent wire rope core (IWRC) cable, and shall be equipped with an end chain and hook. An integral, adjustable, automatic safety brake shall be provided. The winch shall be equipped with a four-way roller guide assembly. The winch shall be equipped with a shaft extension on the curbside with a capstan drive extension (AT&T type).

3.4.22.1 Winch drum guard. A winch drum guard shall be furnished. The guard shall confine the cable to the area between the drum flanges. The guard shall consist of not less than 0.25 inch vertical side plates, conforming to the outside radius of the drum flanges. Six bars,

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0.375 inch by 1.25 inches shall be welded to the vertical side plates. Three bars shall be located on the top and spaced equally on the top radius, and three bars shall be located on the bottom and spaced equally on the bottom radius. The vertical distance between the vertical side plates and the drum flanges shall be not more than half the specified cable diameter.

3.4.22.2 Front bumper. The bumper shall be mounted forward of the winch. The open area on either side of the winch shall be covered with a combination step plate and gravel guard. The bumper shall be either a channel or pipe type. When a pipe type front bumper is furnished, the nominal diameter shall be not less than 3 inches, and shall have a wall thickness of not less than that specified in Schedule 40 of ASTM A120. The pipe type front bumper shall have half-round ball ends.

3.4.22.3 Combination step plate and gravel guard. A combination step plate and gravel guard fabricated of not less than 14 gage (0.0747 inch) steel tread plate exclusive of the projections shall be furnished. The step plate shall be secured to the front bumper and shall be not less than 0.25 inch and not more than 0.375 inch from the chassis sheet metal. The step plate shall be capable of supporting 300 pounds per square foot. The step plate shall not deflect more than 0.1250 inch under the loads imposed.

3.4.22.4 Winch reel. A reel for payout and takeup shall be provided. The reel shall be designed to operate from the end of the extended winch shaft. Reel yokes shall be controlled by a handle at the end of the reel shaft. The spindle shall be fitted with a bayonet type socket for attachment to the shaft. Reel spiders and yokes shall be made from heat treated aluminum. The inside dimension of the coil shall be approximately 20 inches, maximum outside diameter approximately 28 inches, thickness or width of the coil approximately 6 inches to 9 inches.

3.5 Body assembly. A line maintenance and construction truck body for heavy duty work shall be furnished. The body assembly shall include cabinets on each side of a center loadspace and a hydraulically operated turret and derrick. For type II vehicles, racks above the cabinets and a telescoping roof shall be furnished. Unless otherwise specified, a nonconductive derrick extension and aerial basket(s), and other equipment as specified herein, shall be furnished. The body shall have the minimum dimensions specified in tables VI and VII. The turret for type I shall be located at the rear of cab. The turret for type II shall be located at the right rear corner of the body. The turret for type III shall be located over the rear axle. The cabinet space shall not be utilized to accommodate outriggers or access steps, except minor cabinet cutouts may be made to accommodate outriggers. Cutouts shall be covered with sheet metal to present a completely enclosed and weatherproof cabinet interior. Safety tread steel plate shall be installed in the loadspace and, for type I and III vehicle, on top of the cabinets. The floor of the loadspace shall be capable of supporting a live load of 500 pounds on any square foot of area without permanent deformation of the floor.

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TABLE VI. Type I and III dimensions (inches).

Feature	Size 55		Size 70	
	Model 4x2 and 4x4	Model 6x4 and 6x6	Model 4x2 and 4x4	Model 6x4 and 6x6
* Rear-of-cab to rear axle	101	101	101	101
Length of body, exclusive of the derrick and platform extension	118	144	137	168
Cabinet lengths, front to rear	18, 18, 59, 24	18, 18, 54, 54	24, 30, 59, 24	18, 18, 54, 54, 24

TABLE VII. Type II dimensions (inches).

Feature	Size 55 Model 4x2 and 4x4	Size 70 Model 4x2 and 4x4	Size 70 Model 6x4
* Rear-of-cab to rear axle	101	101	101
Length of body, exclusive of the derrick and platform extension	130	118	144
Cabinet lengths, front to rear	24, 24, 24, 59	24, 18, 18, 59	18, 18, 54, 54
Racks over cabinets:			
Length	130	118	144
Height	26	26	26
Width	10	10	10

3.5.1 Bumper and platform extension. A combination bumper and platform extension shall be provided. A safety tread steel plate shall be provided on top of the platform extension. The combination bumper and platform extension shall be capable of storing the tamper, pole jack carrier, and the highway warning kit in accessible locations.

3.5.1.1 Stirrup steps. Stirrup steps shall be furnished on the combination bumper and platform extension. A grab handle shall be installed to assist personnel in entering the body loadspace.

3.5.1.2 Access steps. Access steps and grab handles shall be furnished on each side of type I and type III vehicles. The access steps shall be between the body and the chassis cab and shall provide for entrance and exit to and from the turret control console or dual controls, as applicable, or, for type III, for access to the body loadspace. The steps shall be provided with safety tread steel plate.

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3.5.2 Cabinets. Cabinets, partitions, and shelves shall be constructed of not less than 20 gage (0.0359 inch) high tensile steel or 18 gage (0.0478 inch) carbon steel, single thickness, reinforced, or 22 gage (0.0299 inch) carbon steel, double panel, reinforced. The cabinet doors shall be double panel, reinforced construction of not less than 22 gage (0.0299 inch) sheet steel. The doors shall be weatherproof and constructed to maintain fit and alinement in service. The doors shall have continuous rod or hem type, heavy duty hinges, with stainless steel hinge pin and stainless steel, nylon, or bronze insert sockets. Latches shall have recessed-handles. Cylinder type locks, keyed alike, shall be installed in each door. Moving parts of latches and locks shall be constructed of or plated with corrosion-resistant material. The cabinet doors over 44 inches in height or length shall have pillar bolts or two latches. Horizontal doors shall have corrosion-resistant stay chains and shall be reinforced for use as shelves capable of supporting a live load of not less than 25 pounds on any square foot of area. The cabinets and shelves shall be capable of supporting a live load of 100 pounds on any square foot of area. Rubber fenders shall be installed around the rear wheelwells, extending to at least the outside of the tires.

3.5.2.1 Cabinet arrangement. Vertical cabinets shall have a minimum length of 18, 24, or 30 inches and horizontal cabinets shall have a minimum length of 54 or 59 inches arranged as shown in tables VI and VII and as follows:

(a) Full height front vertical cabinets shall be a minimum of 15 inches in depth, and 44 inches in height and shall be equipped with bolted on or screwed on rope hooks
(b) Full height rear vertical cabinets shall be a minimum of 15 inches in depth and 45 inches in height
(c) Intermediate vertical cabinets shall be a minimum of 15 inches in depth
(d) Horizontal cabinets shall be located over the wheelwell housings and shall be a minimum of 15 inches in depth, 20 inches in height and 59 inches in length for model 4x2 and model 4x4, and 54 inches in length for model 6x4 and 6x6
(e) Horizontal cabinets shall be equipped with small parts bins with adjustable spacers, shelves, and partitions and a rubber goods compartment which shall be insulated and vented. Vertical cabinets shall be equipped with not less than two adjustable tray type shelves. A water cask, when required (see 3.5.6), shall be mounted in the curbside vertical compartment. An insulated hot stick compartment for accommodating hot sticks at least 10 feet in length shall be furnished in either the streetside or curbside cabinets.

3.5.2.2 Racks, type II. Type II vehicles shall be furnished with equipment racks. The equipment racks and shelves shall be provided above the cabinets, and shall open to the center aisle. Swivel type hooks shall be provided on each side of the aisle. Outside holders for six pike-poles shall be provided on one side and ladder hooks with retainers shall be provided on the other side.

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3.5.2.3 Roof type II. A telescopic sheet metal roof over the center loadspace of the vehicle body shall be furnished. The roof shall have one fixed section at the forward end, and two telescopic sections. The roof shall be weathertight.

3.5.3 Lights. In addition to the lighting specified in 3.4.2.3, the following lighting and wiring shall be furnished:

(a) Lights with toggle switches and guards in each cabinet compartment having an exterior door opening
(b) Lights on universal mounts and of sufficient candlepower to illuminate body loadspace and all equipment contained therein and all controls
(c) Two 5.750 inch diameter sealed beam spotlights with universal mounts shall be installed, one on each side of the cab, operable from within the cab
(d) Demountable projecting load light on the hydraulic tilting pole grabber, with a service line and socket when the end of the derrick extends beyond rear of the vehicle (type I only). A receptacle for a light on the rear of the body shall be provided. The light shall be in accordance with Federal Motor Carrier Safety Regulation 393.18
(e) Floodlight mounted on a universal mounting to illuminate the rear of the vehicle and towed vehicle. The floodlight shall be provided with a toggle switch and a switch on the instrument panel
(f) Amber, rotating, vehicular warning light in accordance with MS51317-4. The rotating light shall be mounted with reinforcement at point of attachment on top of the cab on vehicle centerline for 360-degree visibility. The rotating light shall be operated by means of a separate switch with a red indicator light located on the instrument panel and marked to indicate the function of the switch
(g) Wiring for the body shall be in accordance with the applicable portion of SAE J1292. All wiring exposed to possible physical damage such as wiring on a compartment floor, or wiring where tools or materials may cause the wire to be frayed or shorted shall be protected by the use of heavy wall thermoplastic conduit or steel conduit having waterproof connections. Grommets shall be used for all wiring going through any hole or body framework
(h) Domelights with a toggle switch shall be furnished in the type II body.

3.5.4 Crew compartment. When specified for type II vehicles (see 6.2), a crew compartment with seats and backs for four men shall be integrally constructed on the forward end of the body. A personnel door shall be furnished on the right (curb) side. Two side windows and one rear window shall be provided; and a winch observation window shall be installed to enable the operator to observe the winch in operation. A compartment heater and roof vent shall be furnished. A flexible communication tunnel shall also be provided. An additional stowage compartment shall be installed under the floor with an exterior access door on

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the left (street) side if practicable. Interior domelighting shall be provided. The body length and cab-axle dimensions specified in tables VI and VII shall be increased by not less than 39 inches.

3.5.5 Body mounting. A subframe shall be provided. The derrick, outriggers and subframe shall be one integral welded assembly. The subframe shall conform to (a) or (b) below:

(a) The cabinet loadspace floor mounting shall conform to 3.5.5.2 through 3.5.5.2.3. The subframe length shall be at least sufficient to distribute the load evenly on the chassis frame rails for type I, size 55; shall extend from the rear of the chassis frame to at least forward of the rear axle(s) for type II, size 55; and shall be the full length of the chassis frame for types I, II, and III, sizes 55 and 70
(b) The cabinet loadspace floor mounting shall conform to 3.5.5.3 and the subframe shall be the full length of the chassis frame.

3.5.5.1 No welding of chassis frame. No part of the body (cabinets, floor, derrick, turret, outriggers, or subframe) shall be welded to the chassis frame.

3.5.5.2 Flexible mounting for cabinet-floor assembly. The cabinets and the loadspace floor shall be an integral assembly, secured independently of the derrick and the outriggers to the chassis frame or derrick subframe with U-bolts, twin studs or brackets and shall include a wood breaker strip. All but the rearmost cabinet-floor assembly mountings shall be spring loaded to provide for movement of the cabinet-floor assembly independent of the subframe, chassis frame, derrick, turret and outriggers.

3.5.5.2.1 U-bolts or twin studs. When U-bolts or twin studs are used for mounting the cabinet-floor assembly, there shall be not less than four U-bolts or twin studs per side having a 0.5625 inch minimum body diameter with a 0.6250 inch minimum thread diameter. Tie-plates shall be at least 0.50 inch thick. The vehicle chassis frame and the subframe shall be braced, using wood blocks, at each mounting point unless the mounting point is located at a full depth crossmember. Blocks shall incorporate a keeper strap or groove for the mounting bolt and shall be of a width to assure retention. Two tieback straps shall be provided, one bolted to each side of the rear, to maintain cabinet-floor assembly alinement on the chassis frame rails or subframe.

3.5.5.2.2 Brackets. When brackets are used, they shall be bolted to the web of the subframe or of the chassis frame rails. The body mounting brackets shall provide means for drawing down the body on the subframe or chassis rails and provisions shall be made to prevent lateral shifting of the breaker strips. When additional holes are required to secure the mounting

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brackets to the chassis frame rails, they shall be located within the area of the rail which is designated as being safe for drilling in accordance with the chassis manufacturer's body builders layouts. Attachments shall neither interfere with nor obstruct existing chassis components.

3.5.5.2.3 Breaker strips. A hardwood or dense southern yellow pine breaker strip not less than 0.75 inch in finished thickness shall be installed between the longitudinal sills of the cabinet-floor assembly and the subframe. The minimum thickness specified shall be increased as may be required to provide adequate tire chain-to-body clearance. Breaker strips shall have a taper of not less than 0.50 inch in 16 to 18 inches at the forward end.

3.5.5.3 Rigid mounting for cabinet-floor assembly. The cabinets and the loadspace floor, as an assembly or independent of one another, shall be bolted to the full length subframe. The full length subframe and the chassis frame shall be tied together in such a manner as to act as a single member resisting bending and distortion of the cabinets. The section modulus of the full length subframe and the chassis frame combined shall be not less than 70 inches cubed per frame rail. A wood breaker strip conforming to 3.5.5.2.3 shall be provided when the cabinets and the loadspace floor are an integral assembly. When the cabinets are mounted independently of the loadspace floor, wood or metal breaker strips shall be provided when needed to provide relief at rivet heads, bolt heads, or other subframe irregularities.

3.5.6 Water cask. When specified (see 6.2), a standard, commercial, 5-gallon, insulated, potable water cask shall be furnished. The inside liner shall be stainless steel or nontoxic poly material. The outside shell shall be galvanized steel or equivalent. The throat area shall be of sufficient area to permit cleaning of the interior of the water cask. A brass, chrome-plated, or nylon spigot flush mounted to prevent damage shall be furnished. Lugs and bracket(s) shall be provided to mount the cask (see 3.5.2.1(e)). Means shall be provided to retain the water cask and to minimize vibration during road travel. Means shall be provided to drain excess water to the ground. A paper cup dispenser shall be furnished and installed in the cabinet at a location accessible to the water cask.

3.5.7 Telephone maintenance accessories. When a front mounted winch is specified, a collapsible cable reel attachment for the front mounted winch is required by 3.4.22.4. The cable reel shall be provided with a carrier and shall be mounted in the loadspace.

* 3.5.8 Hydraulic driven generator. When specified (see 6.2), a hydraulic driven generator shall be furnished. The generator shall have a power rating of 115 volts and 230 volts, single phase, 3-wire, 60- Hertz, and shall produce not less than 3,500 watts continuous service capacity. The generator and its circuitry shall conform to the requirements of NFPA-70. The generator shall be secured on the left front of the loadspace floor.

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3.5.8.1 Receptacles. Three pairs of 115-volt and 230-volt receptacles shall be provided, one pair at the curbside front, one pair at the streetside front, and one pair at the rear of the vehicle. The receptacles shall be recessed within the body and provided with a weatherproof spring loaded cover.

3.6 Turret and derrick. Turret and derrick structural components shall have a calculated design stress, such that the yield point of the materials used is not exceeded by the induced stress due to the weight of the components plus the stress due to three times the loads specified in 3.3.4. For plastic components, the yield point shall not be exceeded with 6 times the load specified in 3.7.3 in the baskets. The materials used shall meet all applicable performance requirements specified herein.

3.6.1 Hydraulic turret. The vehicle shall be furnished with a hydraulic motor driven turret. The turret shall have a rotating rated torque capacity of not less than 37,000 pound-feet with a safety factor of 3 for Size 70 vehicles and 21,000 pound-feet with a safety factor of 3 for size 55 vehicles. The hydraulic relief valve(s) shall be set to meet the rated loads specified and to provide the minimum specified torque requirements. All journals, bearings, and drive components shall be oil lubricated. A means shall be provided to rotate the turret in the event of hydraulic failure.

* 3.6.2 Derrick assembly. The derrick assembly shall be of the telescopic type, hydraulically extended and retracted by a hydraulic cylinder. The derrick assembly elevation from stowage position to maximum elevation shall be by means of a hydraulic cylinder. The derrick assembly shall be attached to the rotating turret utilizing self-alining bearing(s) and shaft. Under dynamic conditions, the derrick shall support the specified load placed on the upper head sheave in all extended positions without permanent deformation of any components in or along the derrick or supporting structure. The derrick assembly shall be capable of extending and retracting in all selected positions. The derrick shall be equipped with a head sheave and travel rack. The derrick shall be capable of setting poles up to 55 feet for size 55 vehicles and up to 70 feet for size 70 vehicles. Extension and retraction shall be governed by a hydraulic control lever. Rotation shall be governed by a hydraulic control lever, either separately or in combination with extension and retraction. All revolving components shall be furnished with lubrication fittings. All bushings shall be capable of replacement and shall be provided with lubrication fittings.

3.6.2.1 Derrick construction. The derrick shall be designed, engineered, and constructed of materials that display the minimum weight conducive to the configuration and maximum performance requirements specified herein. The hydraulic lines and other associated parts which are fastened to or inside the derrick assembly shall be positioned for easy inspection, ready maintenance, and possible replacement and shall have maximum protection against damage afforded by the configuration of the derrick assembly. The derrick shall be attached to the turret

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by replaceable shafts. The rotating bearing surfaces of the derrick assembly at the turret which are subjected to continuous wear shall be provided with replaceable bearings or bushings. Lubrication fittings shall be provided on all rotating parts. The bearing surfaces shall be adequate to resist torsional moment when inherent with the design.

3.6.2.2 Derrick operation. The rotating turret and derrick assembly shall be hydraulically actuated by separate mechanisms. The derrick operating mechanism shall be capable of limiting the lifting capabilities from rest position to those within the limits of the design stress specified in 3.6 to prevent overloading. For testing purposes, means shall be provided in the hydraulic system to perform the overload tests.

3.6.2.3 Derrick storage. For normal storage, travel and shipment of the vehicle, the derrick shall retract into an essentially horizontal position parallel to the longitudinal centerline of the vehicle. Saddle and tie-down devices shall be provided to support and hold down the derrick assembly to prevent nonactuated raising of the derrick assembly when the vehicle is in transit. Derrick saddles shall be sufficiently padded to prevent damage to any component of the derrick. Type I storage device shall be capable of being swung to the side when not in use to allow a clear opening to the vehicle body loadspace. When the derrick is in the travel position (on type II and III vehicles), the derrick and its components shall not obstruct the vision of the driver.

3.6.2.4 Capacity, height, and radius indicator. An indicator shall be furnished on the operator's side of the derrick to visually display capacities, height and radius.

3.7 Derrick extension and aerial basket. Unless otherwise specified (see 6.2), a manually operated, telescopic, fiberglass, nonconductive extension for the derrick shall be provided. The extension shall mount a one-man fiberglass basket or when specified (see 3.7.3) two one-man fiberglass baskets.

3.7.1 Plastic construction. Extension, basket and components shall consist of extruded, pultruded, molded, or spiral wound fiberglass, and each component shall conform to the performance requirements for electrical insulation specified herein. All components shall be from a combination and multiple layers of sterile electrical grade fiberglass roving cloth or mat with high impact strength resins with weathering and fire-retardant characteristics employed. The exterior surface shall be smooth and of a uniform thickness. When gelcoat is applied, the thickness of gelcoat shall be not less than 0.006 and not more than 0.030 inch. The parts shall be free from uncured areas, gaps, cracks, holes, blisters, fuzzy surfaces, resin pockets, areas lacking resin, laps, wrinkles, delamination, air or gas pockets, patches, porosity, and other similar defects. The material shall be essentially void-free. The entire surface shall be fully cured and free of tackiness. The percentage of glass by weight shall be approximately uniform.

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3.7.2 Dielectric properties. The extension shall withstand a minimum of 100,000 volts ac, 60 Hertz, applied across the full length of the derrick extension for not less than 3 minutes without evidence of dielectric puncture and with leakage of not more than 1.0 milliampere when tested in accordance with ANSI A92.2. Any mechanism within or along the extension shall be made of dielectric materials or so insulated as to demonstrate the same dielectric properties as the extension. The extension components shall be completely insulated from the derrick assembly, turret, and fiberglass basket. The fiberglass basket(s) shall withstand a minimum of 30,000 volts ac, 60 Hertz, without evidence of dielectric puncture. The basket liner(s) shall withstand a minimum of 50,000 volts ac, 60 Hertz for not less than one minute without evidence of dielectric puncture. A decal shall be furnished and installed on the basket stating "NO INSULATION VALUE WITHOUT LINER."

3.7.3 Aerial basket(s). Unless otherwise specified, a one-man, single side-hung basket, having a total payload capacity of not less than 300 pounds, shall be furnished. When specified (see 6.2), two one-man baskets shall be furnished. When two, one-man baskets are furnished the total payload capacity shall be not less than 500 pounds. Minimum interior dimensions of the basket(s) shall be not less than 42 inches in depth, 22.50 inches in width, and 20 inches front to the back. Minimum interior dimensions shall be measured at the bottom of the basket(s) just above the bottom radii. The basket(s) shall be fully enclosed on five sides and provided with a molded-in step, protruding towards the outside, to permit entrance and exit. The basket(s) and mounting bracket shall be accessible for field installation on the derrick extension (see 3.7) and for removal and stowage on the vehicle. Means shall be provided to secure the basket(s) in the vehicle loadspace.

3.7.3.1 Insulated basket liner(s). An insulated basket liner shall be provided for each basket. The liner(s) shall be capable of being easily removed for cleaning and testing their dielectric capacity. The basket liner(s) shall be in accordance with the performance requirements specified in 3.7.2.

3.7.3.2 Flooring. The basket(s) shall have a reinforced floor. No point in the flooring shall deflect more than 1/4 inch nor be permanently deformed when subjected to a vertical load of 300 pounds covering an area of 25 square inches. A nonconductive floor mat shall be provided for each basket.

3.7.3.3 Basket leveling system. An independent basket leveling system shall be provided for keeping the basket horizontal in all derrick positions. The leveling system shall consist of a basket gravity leveling system and shall be independently capable of keeping the basket level with the rated load on the basket with the load positioned on the floor of the basket. An adjustment shall be provided to enable personnel in the basket to control the rate of movement of the basket relative to the derrick position and to lock the basket in a fixed position

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at working height. The leveling system component shall be provided with maximum protection based on design features and safety. The system shall be insulated as specified (see 3.7.2). The leveling system shall operate without breakage in all extended positions that the derrick is capable of attaining.

3.7.3.4 Safety belt(s). A positive-lock type, lineman safety belt with quick-release features shall be provided for each basket. Belt(s) shall have a design strength of 2,000 pounds. The safety belt(s) shall be designed and positioned to allow freedom of movement of the workman in the basket(s). Means shall be provided to attach safety belt fasteners to the end of the extension or basket bracket and to prevent the fastener attachment from being pulled through the plastic wall, in the event of leveling system failure. Fasteners and attachments shall be corrosion-resistant.

3.7.3.5 Basket cover(s). A waterproof basket cover and a means of retaining the cover on the basket while in storage or stowage-in-transit position shall be provided for each basket. Provisions shall be made to store the cover(s) in a compartment of the utility body.

3.7.3.6 Tool tray(s). A fiberglass or polyethylene tool tray, one for each person of basket capacity, shall be provided to fit over the edge of the basket(s). Provisions shall be made to secure and stow the tool tray(s) in a compartment of the utility body.

3.8 Hydraulic digger. A reversible hydraulically driven post-hole digger shall be provided. The motor driven digger shall be capable of boring holes up to 12 feet in depth and up to 24 inches in diameter for size 55, and up to 30 inches in diameter for size 70. The hydraulically operated motor of the digger shall be capable of developing, at the output shaft, not less than 7,000 foot-pounds of torque at a shaft speed of 34 revolutions per minute (rpm). For high speed spin-off of dirt, the digger shall rotate at not less than 75 rpm. The digger shall be equipped with a boring head similar or equal to the Pengo boring heads. Hydraulic flow control valves shall be furnished to prevent overspeeding the digger motor.

3.8.1 Augers. When specified (see 6.2), size 55 vehicle shall be furnished with 12-inch, 18-inch, and 24-inch augers. When specified (see 6.2), size 70 vehicle shall be furnished with 12-inch, 18-inch, 24-inch, and 30-inch augers. The augers shall suit the boring head specified in 3.8. Each auger shall be furnished with a pilot bit for center drilling the hole and the required number of auger teeth. Auger teeth shall be capable of boring holes in swamp, loose sand, hard terrain, and rocky soil. Stowage for shipping of the augers shall be provided on the vehicle.

3.8.2 Spare auger teeth. One complete set of spare auger teeth shall be furnished for each 12-inch, 18-inch, 24-inch, and 30-inch auger furnished.

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3.8.3 Digger and auger stowage on derrick assembly. Stowage for the digger with up to 24-inch size auger in place shall be provided on the derrick assembly. A safety device to prevent damage to the auger storage bracket, when winding a cable to stow the auger, shall be furnished. A hydraulic or mechanically operated automatic locking device with a positive lock to stow the digger and auger shall be furnished. The stowing mechanism shall raise the digger and auger into a position along side of derrick assembly and place into the automatic locking device to retain and release the digger and auger as required. A slip clutch or other torque limiting device shall be provided for the winding cable. A safety check valve shall be furnished to prevent the clutch and digger from slipping. Stowage provisions shall have anti-bounce provisions for the digger and auger.

3.9 Derrick attachments.

3.9.1 Hydraulic pole grabber. A hydraulically tilted pole grabber shall be furnished. The pole grabber shall be located at the end of the derrick assembly. The assembly shall be capable of setting 55-foot length poles for size 55 and 70-foot lengths for size 70. The pole grabber assembly shall consist of two steel arms actuated by a hydraulic cylinder. The assembly shall be capable of tilting not less than 80 degrees. The pole grabber shall be provided with automatic safety pilot check valve(s).

3.9.2 Derrick winch. A hydraulic, motor driven derrick winch shall be mounted at the base of the derrick or on the turret assembly. The winch shall conform to SAE J706 and shall be equipped with an automatic brake. The winch drum diameter shall be not less than 8 inches. The winch shall be provided with a hydraulic relief valve set to restrict the line pull to the winch rated capacity. The winch shall have a rated single line pull capacity on the bare drum of not less than 12,000 pounds for size 55 and 15,000 pounds for size 70. The winch shall be provided with a cable tensioner that resists cable overlapping on the drum.

3.9.2.1 Winch rope. Unless otherwise specified, the winch shall be provided with not less than 118 feet of 1.1250 inch diameter polyester double braid rope. The rope shall have an average breaking strength of not less than 46,000 pounds. The rope shall be coated for maximum abrasion resistance with a material that does not prevent splicing and does not interfere with dielectric properties. Eye splices shall be provided at both ends of the rope. A removable thimble shall be provided at the free end. A swivel hook with down haul weight shall be provided. When specified (see 6.2), if an extension and a basket are not required, not less than 150 feet of 0.5625 inch wire rope shall be provided in lieu of the polyester rope. The winch wire rope shall be improved plow steel, 6x37, hemp core, preformed, regular lay, with a cable loop formed at the end.

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3.9.2.2 Cable guards. Two hinged cable roller guards shall be furnished on top of the derrick boom, one near the front, the other near the rear. The guards shall retain the cable and allow up and down movement of the cable. Rollers shall be furnished above and below the cable.

3.10 Derrick controls. A complete set of hand-operated controls for the hydraulic turret, derrick assembly, winch, digger, and pole grabber shall be provided. All gages, controls, and switches furnished shall be mounted to resist shock and shall be weatherproof. The centerline of the controls shall be at least 4 inches apart. The controls shall be self-centering and shall revert to a neutral position when released by the operator. A safety interlock system shall be provided to prevent operation of the derrick assembly when the outriggers are not extended. A means shall be provided to retract the derrick assembly and rotate the turret into stowage position, and retract the outriggers in the event of hydraulic failure. The switches, instruments, and pressure gages on the control station shall be panel mounted and identified as to their function and installed in a manner to facilitate removal and servicing. Gages and instruments shall be provided and shall be waterproof and weatherproof. Identification markings shall be on photo-etched, corrosion-resistant metal plates. At least the following gages and controls shall be furnished on the console or the dual controls for type I, on the console for type III, and in the left rear compartment control cabinet for type II:

(a) Key-operated 3-wire, remotely operated start and stop switch
(b) Foot operated engine throttle
(c) Tachometer for each control location
(d) Dash light(s) with rotating shields and toggle switch to illuminate control panels
(e) Pressure and temperature gages for the hydraulic system
(f) Restricted return line filter warning light.

3.10.1 Controls on rotating turret, type I and type III. Unless dual side controls are specified for type I (see 3.10.2), controls for type I and type III shall be furnished on a control console on the rotating derrick. An operator's seat shall be furnished. Access to the control console shall be provided as specified in 3.5.1.2. The control console shall be weatherproof. The operator's platform shall be furnished with safety tread steel plate or grating. A weatherproof cover shall be provided to cover the controls. Means shall be provided to retain the cover.

3.10.2 Dual side controls, type I. When specified (see 6.2), dual controls for type I shall be provided on each side of the turret. The controls shall be identified as to their function. The identification plates shall be located at the controls. Access to controls shall be provided as specified in 3.5.1.2. The control panels shall be weatherproof. The operator's platform shall be furnished with safety tread steel plate or grating. A weatherproof cover shall be provided to cover the controls. Means shall be provided to retain the cover.

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3.10.3 Left rear controls, type II. The controls for type II vehicles shall be located at the left rear of the vehicle. An operator's platform shall be furnished. The platform shall be located at the left rear corner of the body. The platform shall isolate the operator from ground in the event of the derrick contacting an energized electrical line. The platform shall be of the folding type.

3.10.4 Controls in basket. In addition to the lower controls, controls shall be provided at the basket(s) for rotating, extending, retracting, elevating and lowering the derrick to its fullest extent. If telescopic linkage type controls between the basket(s) and the lower controls are provided, the top section of the telescopic tubes shall be spiral wound fiberglass. Design and placement of the controls shall be so as to prevent inadvertent actuation by the operator. The lower controls shall be capable of overriding the basket controls and of rendering the basket controls inoperative. When two baskets are furnished, the controls in the basket shall be within easy reach of either basket. The controls at the basket shall be located at the right or left side of the basket in a position easily accessible to the operator facing the work area. Controls on the front or rear of the basket shall not be furnished. The controls shall be identified as to their function. When two baskets are furnished, the identification plates shall be located to be seen from either basket. The identification plates shall be located at the controls and shall be readable in all positions of the extended derrick.

3.11 Outriggers. Two hydraulically operated outriggers shall be provided for size 55 and four hydraulically operated outriggers shall be provided for size 70 vehicles. The outriggers shall enable leveling of the vehicle on inclined surfaces and shall ensure adequate stability with relation to the ground while operating the derrick under loads and conditions specified in 3.3.3 through 3.3.8. The outrigger cylinder hydraulic fittings, hoses, tubing, and pilot check valves, when not enclosed, shall be protected by metal guards and accessible for service. The guards shall withstand being stepped on without damage to the guards, hydraulic lines, or fittings. The outriggers shall be designed so that they may be extended over curbings and sidewalks.

3.11.1 Outrigger controls. An individual control for each outrigger shall be furnished. All controls for the left outriggers shall be provided at the rear left corner of the vehicle and controls for the right outriggers shall be provided at the rear right corner of the vehicle. All outrigger controls shall be installed in a readily accessible location which permits operation from ground level. An automatic pilot check valve shall be provided for each outrigger to block the hydraulic fluid in the cylinder and to prevent movement of the outriggers in the event of a damaged hydraulic line. Interlocks shall be furnished to prevent the operation of the derrick assembly until the right and left side outriggers have descended into place on the ground surface.

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Interlocks shall not interfere with operation of the hydraulic tool circuits, with the outriggers up or down. A self-actuating locking device shall prevent accidental raising of the outriggers while the derrick assembly is elevated from stowage position. Each outrigger shall be provided with an automatic, positive, spring actuated lock, or with an automatic pilot check valve which holds the outrigger in the retracted position without settling and which may be manually released.

3.11.2 Outrigger movement. The outriggers for type I and type III shall be transversely spaced not less than 11 feet, 6 inches, measured at the ground pad, and shall be of the hydraulically operated type. The outriggers for type II shall consist of a hydraulically operated slant type telescoping outrigger on the streetside rear and a hydraulically operated transverse type outrigger on the curbside rear. The streetside outrigger shall extend not less than 54 inches, measured from the longitudinal centerline of the vehicle. The curbside outrigger shall extend not less than 76 inches, measured from the longitudinal centerline of the vehicle. For type II, forward outriggers shall be transversely spaced not less than 11 feet, 6 inches, measured at the ground pad and shall be of the hydraulically operated type. All transverse measurements shall be made to the pad hinge pin or to the center of the pad, at the ground, with the vehicle resting on the tires, on level ground.

3.11.3 Outrigger pads. Each outrigger ground pad shall have a ground contact surface area of not less than 196 square inches. Ground pads shall be of the hinged or fixed type.

3.11.4 Outrigger warning light. A red warning light shall be installed in the cab in a conspicuous location to warn the driver that the outriggers are extended.

3.12 Hydraulic system. The hydraulic system shall consist of the components required to perform all hydraulically actuated operations of the derrick assembly, winch(es), components, accessories, and outriggers with the degree of control specified herein. The mechanisms and fluid reservoirs shall be positioned or shielded to prevent hazardous contact with heated surfaces or electrical apparatus. An electric pushbutton throttle control shall be furnished. The electric pushbutton shall activate the engine throttle to increase the engine speed to that required for rotation of the derrick within the speed range specified in 3.3.5 and for operation of the outriggers, pole puller and tamper, and to return the engine to idle speed. An accumulator may be used as a component of the hydraulic system to provide the maximum degree of temperature control under continuous service.

3.12.1 Hydraulic cylinders. All hydraulic cylinders shall incorporate seals to prevent entrance of moisture and dirt into the cylinders, and to prevent leaks of hydraulic fluid from the cylinders. Seals applied to reciprocating parts shall be of a design and type proven in hydraulic applications. All hydraulic cylinders for actuating the derrick and outriggers shall be provided

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with pilot check or holding valves to prevent descent of the derrick assembly and retraction of the outriggers because of loss of hydraulic pressure. Means shall be provided to install pressure gages for testing the hydraulic system pressure and relief settings. Hydraulic ram piston rods shall be chrome-plated full length with sufficient thickness to withstand a 30-hour salt spray test in accordance with ASTM B117.

3.12.2 Hydraulic lines and fittings. Hydraulic lines shall be located to facilitate maximum ease of removal and installation, visual inspection for failures, and maximum protection from damage during operation of the derrick assembly. Flexible hydraulic hoses or collector rings shall be used where flexing of hydraulic lines is encountered and where installation of components requires a flexible coupling. Standard straight-thread, flared type fittings or adaptions to flared type fittings or tapered pipe thread with dry seal fits shall be furnished. The hydraulic lines to the digger shall be of the telescopic or cat-track type to prevent damage that is encountered during operation of the equipment. The hydraulic lines and fittings shall be air- and liquid-tight and shall be installed to prevent vibration and chafing. The system shall be designed to prevent hydraulic transients. Provisions shall be made for bleeding air from the hydraulic system and for completely draining all hydraulic fluid lines. A manual type bleeder shall also be provided as on-vehicle equipment on closed-center systems. The hose assemblies and all seamless steel hydraulic lines shall be adequate to provide the rate of flow, temperature and pressures required to operate the system smoothly and efficiently. Galvanized pipe and copper tubing shall not be furnished. A pressure gage shall be provided on the discharge side of the hydraulic pump.

3.12.3 Hydraulic pump system. A hydraulic pump with necessary pressure and flow capacities to operate all hydraulic powered components shall be provided. The hydraulic pump shall be driven by a full torque, heavy duty power takeoff unit attached to the truck chassis transmission. Chain drives shall not be used. The pumping system shall include an adjustable pressure relief valve or unloading valve with a range of adjustment which allows raising or lowering of the derrick assembly with any working or test loads specified herein.

3.12.4 Hydraulic fluid reservoir. A hydraulic fluid reservoir shall be provided and shall have sufficient capacity to hold the required volume of hydraulic fluid, with not less than 10 percent for expansion space. The reservoir shall be fully baffled. The reservoir shall be provided with a filler opening and removable oil strainer and a vented filler cap accessibly located to facilitate filling the tank from a one-gallon container using a funnel. The vented filler cap shall have means to prevent fluid from splashing out of the tank during movement of the vehicle. The filler cap shall be attached to the tank by a corrosion-resistant chain to prevent loss. Provision shall be made for inspection and cleaning of the interior and draining of the reservoir. The reservoir drain shall be located to prevent fluid from draining on other

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components of the vehicle and to allow recovery of the fluid in a container. The reservoir shall have a corrosion-resistant metal plate indicating the type of hydraulic fluid required and the maximum fluid level in the reservoir with the derrick in travel position and with the hydraulic lines and components full of fluid. A sight level or dip stick gage shall be furnished to indicate the level of the fluid. The tank shall be clean.

3.12.5 Hydraulic fluid. Unless otherwise specified, hydraulic fluid conforming to the manufacturer's standard or recommended requirements shall be provided in the hydraulic system. When specified (see 6.2), hydraulic fluid conforming to MIL-H-83282 shall be provided in the hydraulic system.

3.12.6 Oil filter and strainers. A full flow type 10-micron discharge oil filter in the return line to the reservoir shall be provided. A 100-mesh suction line strainer in the hydraulic reservoir shall be provided. The filter and the suction strainer shall be easily accessible for periodic cleaning or for replacement of the filter cartridge. The filter and the strainer shall have sufficient capacity to allow the maximum rated pump discharge.

* 3.12.7 Hydraulic fluid temperature. The hydraulic fluid temperature shall be maintained within safe limits to prevent seal and valve damage and to prevent approach to flash-point of the fluid during all operating conditions of the vehicle, including use of the derrick assembly for long periods of time. The hydraulic fluid temperature shall not exceed a maximum of 180ºF in a maximum ambient temperature of 125°F. Provisions shall be made to prevent malfunction or unactuated operation of the hydraulic control system due to changes in fluid temperature.

3.13 Ground hydraulic tool outlet. A hydraulic tool outlet shall be provided in a curbside cabinet or in a protected position on the rear or under the platform. When specified (see 6.2), hydraulic tool outlets shall be provided on the curbside and streetside of the vehicle. The discharge port(s) and return port(s) shall be provided with corrosion-resistant caps, secured with corrosion-resistant chains to protect the threads and the valve from contamination. Flow divider control valves shall use the hydraulic fluid from the hydraulic system of the vehicle. Unless otherwise specified herein, one tool circuit with quick disconnect, dripless couplers shall be furnished for attaching the pressure and return hose lines for the motor driven hydraulic tools. The oil flow for the tools shall be regulated by control valve(s) to meet the flow requirements of the hydraulic tools. The maximum pressure shall be 1,700 pounds per square inch (psi) to 1,800 psi and the back pressure in the return line shall not exceed 150 psi.

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3.13.1 Ground tool hose. Two 50-foot lengths of hydraulic wire braid hose with quick disconnect, dripless couplers on both ends with caps for use on the ground level shall be furnished. A warning shall be provided in the instruction plates (see 3.1.1.1.2) that the wire braid hydraulic hoses shall not be used on or near electrical lines. The hose shall be furnished with dust caps. The hose fittings shall be compatible with the tool outlet(s) specified in 3.13 and with the tools specified in 3.13.2 through 3.13.5.

3.13.2 Hydraulic pole puller and jack. A hydraulic pole puller and a hydraulic jack shall be furnished. The pole puller shall have the capacity to lift poles up to 55 feet in height for size 55 and 70 feet in height for size 70, when the pole is in the ground to a depth of 12 feet. The pole puller and jack shall have a lifting force of not less than 40,000 pounds and shall be capable of vertically lifting the pole not less than 15 inches per cycle. A pole pulling chain, base plate, and controls shall be provided. The jack shall be operated from the outlet in the curbside compartment specified in 3.13. The hoses specified in 3.13.1 shall be utilized. Means shall be provided to secure the pole puller and jack on the vehicle.

3.13.3 Hydraulic tamper. A hydraulically operated tamper shall be furnished. The tamper shall be capable of tamping the earth around a pole when the pole is 12 feet in the ground. The tamper shall be capable of delivering a force of not less than 25 pounds at not less than 1,200 impacts per minute. A hydraulic control shall be furnished. The tamper shall be operated from the ground tool outlet specified in 3.13 using the hydraulic hoses specified in 3.13.1. Provisions shall be made for stowage and securing of the tamper in a body cabinet compartment.

3.13.4 Hydraulic capstan hoist. When specified (see 6.2), a hydraulic capstan hoist shall be furnished. The hoist shall be equipped with quick disconnect, dripless fittings and shall utilize hoses provided (see 3.13.1) and hydraulic power from the tool outlet (see 3.13). The drum shall be one direction drive and shall have a backstop on the output shaft to prevent reverse travel. Lowering will be accomplished by letting the rope slip on the drum. The capstan hoist shall be complete with a link chain and screw clamp attachment device for mounting to surfaces other than on a round pole. The hoist shall have a rated single line pull capacity of not less than 1,250 pounds on the bare drum. The hoist shall have a remote control, foot operated, open-center valve. Suitable stowage area in the body shall be provided.

* 3.13.5 Hydraulic sump pump. When specified (see 6.2), a hydraulically driven, submersible sump pump shall be furnished. The pump output capacity shall be not less than 300 gallons per minute (gpm) against a 10-foot head when operating at the hydraulic flow provided by the system specified (see 3.13). The pump size shall permit placing the strainer in a 7-inch diameter hole. The pump weight shall be not more than 25 pounds. The pump shall be provided with not less than a 6-inch diameter strainer. The pump shall be provided with not less

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than a 2-inch inlet. The strainer and pump shall be capable of passing solids up to 0.3750 inch in diameter. The outlet size shall be not less than a nominal 2 inch size. The pump shall be capable of running dry without damage. Not less than 25 feet of collapsible discharge hose shall be supplied with the pump. The hydraulic hose specified in 3.13.1 shall be utilized to drive the pump. Suitable stowage provisions in the body for the pump and hose shall be provided.

3.14 Basket hydraulic tool outlet. Unless otherwise specified (see 6.2), a hydraulic tool outlet shall be furnished at the basket(s) by means of a dual hose and a hose reel or other automatic hose stowing arrangement. The hose reel shall be mounted in an accessible location on the turret or derrick. The hose shall automatically extend and retract with movement of the derrick. The hose and reel shall be capable of withstanding 1,800 psi of line pressure and of meeting the dielectric properties specified in 3.7.2. The hose shall have quick disconnect, dripless type fittings at the basket outlet. A weatherproof cover shall be provided for the hose reel. The following shall be included:

(a) Flow control valve
(b) Line check valve
(c) Roller hose guides.

3.14.1 Basket tool hose. When a basket hydraulic tool outlet is required, an 8-foot dual hydraulic hose, for use in the basket, shall be furnished. The hose shall be rubber, nonconductive type with a compatible quick-disconnect, dripless fitting on each end. The hose ends shall be furnished with dust caps.

3.14.2 Basket hydraulic chain saw. When specified (see 6.2), a hydraulic chain saw shall be furnished. The saw shall be equipped with quick-disconnect, dripless fittings. The saw shall have a chipper type chain and shall be suitable for cutting trees up to 15 inches in diameter. A suitable stowage area in the body shall be provided.

3.14.3 Basket hydraulic chain saw, pole mounted. When specified (see 6.2), a hydraulic chain saw, pole mounted, shall be furnished. The chain saw shall be lightweight (approximately 10 pounds) and shall have a cutting capacity of 9 inches. The saw shall be equipped with quick disconnect, dripless fittings. The pole shall be approximately 8 feet long and shall have a fiberglass type of body to provide good insulation. A suitable stowage area in the body shall be provided.

3.14.4 Basket hydraulic circular saw, pole mounted. When specified (see 6.2), a hydraulic circular saw, pole mounted, shall be furnished. The saw shall be equipped with quick-disconnect, dripless fittings. The saw shall have a capacity of not less than 3.25 inches and shall operate at approximately 8,900 feet per minute. The pole shall be approximately

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6.50 to 7 feet long and shall have a fiberglass type of body to provide good insulation. The saw shall have a 44-tooth blade. A suitable stowage area in the body shall be provided.

3.14.5 Basket hydraulic impact wrench and drill. When specified (see 6.2), a hydraulic impact wrench and drill shall be furnished. This combination tool shall be reversible and capable of installing and removing lag screws and other hardware. The wrench shall have a 0.6250 standard bolt size capacity with a 0.50 inch square drive. The tool shall be capable of developing not less than 200 foot-pounds of torque with at least 1,350 impacts per minute and shall be equipped with a torque regulating device. A quick change drill chuck shall be furnished. Quick-disconnect, dripless fittings shall be provided. A suitable stowage area in the body shall be provided.

3.14.6 Basket hydraulic pruner, pole mounted. When specified (see 6.2), a hydraulic pruner, pole mounted, shall be furnished. The pruner shall be equipped with quick-disconnect, dripless fittings. The pruner shall have a capacity of 2 inches in diameter. It shall be mounted on a fiberglass type pole to provide insulation and shall be approximately 7 feet long. A suitable stowage area in the body shall be provided.

3.15 Auxiliary engine driven air compressor. When specified (see 6.2), an auxiliary engine driven air compressor shall be furnished.

* 3.15.1 Auxiliary engine. An auxiliary air cooled engine shall be provided to drive the compressor. The auxiliary engine shall be a two or four-cycle diesel engine on a truck chassis with a diesel engine or a four-cycle gasoline engine on a truck chassis with a gasoline engine. The engine shall be mounted on the body platform in conjunction with the air compressor. The mounting shall minimize transfer of vibration. During normal operating conditions, the continuous-duty horsepower output required by all components and accessories driven by the engine shall not exceed 75 percent of the full throttle rating of the engine. The size and weight of the engine assembly shall be held to a minimum consistent with the performance requirements specified herein. When a gasoline engine is furnished it shall be capable of warranted operation on unleaded fuel.

3.15.1.1 Engine components. The auxiliary engine shall be equipped with at least the following components:

(a) Magneto or distributor type ignition (gasoline engine only)
(b) Self-starter, 12-volt
(c) Resistor-type spark plugs (gasoline engine only)
(d) Air cleaner
(e) Choke (gasoline engine only)
(f) When power plant heaters are specified (see 3.4.1.7), a 300-watt immersion type heater with 170°F to 195°F thermostat shall be installed in the oil pan through any convenient opening

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(g) An engine hour meter, having a totalizing mechanism of not less than 9,999 hours, shall be provided.

3.15.1.2 Engine control panel. The engine accelerator, key-operated ignition (starting) switch, and palm button starter shall be installed in a weather- and waterproof cabinet on a control panel, and each shall be identified as to its function. The control panel shall be furnished with a side-hinged door, spring-loaded latch, with hasp for locking with a 1-inch padlock.

3.15.1.3 Electrical system. The electrical system shall utilize the truck-chassis battery and shall be in accordance with Federal Motor Carrier Safety Regulations 393.30, 393.31 and 393.33.

3.15.1.4 Fuel pump. A fuel pump shall be furnished on the engine. When lifting fuel against a head of 4 feet, the pump shall have sufficient capacity to supply the engine at its maximum rated load. An in-line fuel filter shall be provided.

3.15.1.5 Governor. A variable-speed governor shall be furnished. The governor shall automatically control the engine speed at the selected rated speed at all fluid pressures and flows required. A manually-operated hand throttle shall control the engine at all speeds from the manufacturer's recommended minimum governed speed up to the maximum operating speed.

3.15.1.6 Auxiliary engine fuel supply. Unless an auxiliary engine fuel tank is specified, the auxiliary engine shall utilize the truck chassis engine fuel tank(s). A shut-off valve shall be provided in the fuel line and shall be located in an accessible location. The fuel system shall be in accordance with Federal Motor Carrier Safety Regulations 393.65 and 393.67.

3.15.1.7 Auxiliary engine fuel tank. When specified (see 6.2), an auxiliary engine fuel tank shall be furnished. The fuel tank shall have a capacity to provide fuel for at least 4 hours of full-load operation. The fuel tank cap shall be attached to the tank by means of a corrosion resistant chain. A sump shall be provided in the fuel tank suitable for collection and accumulation of water. A drain shall be so located at the bottom of the sump for complete drainage of the fuel tank. Drainage shall be provided to prevent spillage of fuel on any area of the vehicle. The tank filler shall be provided with a protective screen and vent. A shut-off valve shall be provided in the fuel line, adjacent to the fuel tank. A corrosion-resistant metal plate shall be furnished and marked denoting the type of fuel to be used and shall be located in a conspicuous location near the filler cap. The fuel line(s) shall be in accordance with Federal Motor Carrier Safety Regulations 393.65 and 393.67.

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3.15.1.8 Exhaust system. The exhaust system shall conform to Federal Motor Carrier Safety Regulation 393.83 and shall be furnished with a spark arresting muffler. Means shall be provided to vent the exhaust fumes away from the cab and body. The spark arrester shall have an 80 percent arresting efficiency when rated in accordance with SAE J350. The exhaust system or the hydraulic system shall be shielded so that, in the event of a hydraulic line rupture or hydraulic fluid leak, the hydraulic fluid will not come into contact with any part of the (auxiliary) engine exhaust system.

3.15.2 Air compressor. The air compressor shall be air-cooled, of the two-stage type with constant speed control; shall have an actual displacement of 30 cubic feet per minute; and shall provide a pressure of not less than 150 pounds per square inch (psi). The air compressor shall be complete with receiver, moisture separator, automatic oilers, shock- and waterproof pressure gage, safety valve, and all necessary equipment for satisfactory operation of the compressor. The receiver tank shall have a capacity of not less than 30 gallons. An air pressure regulator having a setting range of at least 50 psi to 150 psi shall be furnished. The air compressor shall be powered by the auxiliary engine and so mounted as to minimize transfer of vibration.

3.15.2.1 Air outlets and lines. An air line extending from the compressor to a quick disconnect type outlet at the curbside of the vehicle shall be provided. A quick-disconnect type tool outlet shall be furnished in the basket and shall be connected to the compressor by means of a hose and hose reel. The hose reel shall be mounted in an accessible location on the turret or derrick. The hose shall automatically extend or retract with the movement of the derrick. The hose shall meet the dielectric properties specified in 3.7.2. A four-way roller guide for the air hose shall be provided.

3.16 Rear winch. When specified (see 6.2), a rear mounted, hydraulic planetary or hydraulic helical gear winch shall be furnished. The winch shall have a rated single line pull capacity of not less than 15,000 pounds on the bare drum. The winch shall be provided with a hydraulic relief valve set to restrict the line pull to the winch rated capacity. The winch shall be of the single drum type and shall conform to SAE J706. The winch assembly shall be equipped with an automatic cable level wind, a cable tensioner, a shaft extension (AT&T type) on the curbside, a two- or three-speed transmission, and a hydraulic motor. The drive for the shaft extension (AT&T type) shall be located either on or off the winch assembly. An integral, adjustable, oil cooled, automatic safety brake shall be furnished. The winch shall have an operating efficiency to endure 8 hours of continuous operation under load without malfunction of any component in the winch. The winch shall be suitable for use in pulling underground cable and for use with the derrick. The winch shall be provided with guards over the chain drive, the sprockets, the cable tensioner and the cable. The winch shall be mounted directly to the chassis frame at the forward end of the body. The winch controls shall be furnished at the rear of the

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body, in a location providing the operator with a clear view of the winch line as it passes through the swivel pulley and to the rear of the vehicle. The winch controls shall be located at least three feet from the swivel pulley in a location minimizing the possibility of operator injury should the winch line break.

3.16.1 Cable. Not less than 1,000 feet of cable shall be furnished on the rear winch. The cable shall be 0.50 inch, improved plow steel, 6x19, hemp core, preformed, regular lay, with a cable loop formed at the end. The winch cable shall be threaded through a tunnel under the body floor.

3.16.2 Universal sheave. A universal swivel sheave, for threading the winch cable, shall be provided at the rear of the body. The universal sheave shall have not less than a 30,000-pound capacity rating. The sheave section of the assembly shall swivel on double row ball bearings. A spring tension guide head shall be furnished to maintain line position.

3.17 Safety. All equipment operating at extreme temperatures and all rotating or reciprocating parts, which are of such a nature or so located as to become a hazard to operating personnel, shall be insulated, fully-enclosed, or properly guarded.

3.18 Dissimilar metals. All dissimilar metals throughout the vehicle shall be insulated from one another to prevent galvanic or electrolytic action.

3.19 Lubrication system. The lubrication system shall be furnished with fittings for all bearings and other working parts which require lubrication for continued service. The vehicles shall be lubricated, prior to delivery, with an applicable grade of standard lubricant for the ambient air temperature at the delivery point.

3.20 Servicing and adjusting. Prior to acceptance of each vehicle by the Government inspector, the contractor shall service and adjust each vehicle and its mounted equipment for operational use including at least the following: cycling the entire turret and derrick hydraulic system; alinement of lights; adjustment of the engine; adjustment of electrical and brake systems; filling and charging of batteries; alinement of front wheels; inflation of all tires; complete lubrication of chassis, engine, running gear, and mounted equipment with grades of lubricants and hydraulic fluids recommended for the ambient air temperature at the delivery point; servicing of the cooling system with a solution of ethylene glycol type antifreeze and water in equal parts by volume; and servicing of the windshield washer reservoir with water and appropriate additives.

3.21 Workmanship. The quality and workmanship of all component parts, including fabricated parts, machined parts, and castings, shall be commensurate with sound manufacturing standards for similar equipment. The vehicles shall be free from all burrs, sharp edges, die, tool

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and metal handling marks, rust, scale, and cracks. All metal and plastic materials used shall be free from delamination defects, and shall be prime material. Defective components or parts and assemblies which have been modified to overcome deficiencies shall not be furnished.

3.21.1 Welding. Adequate preparation of metal prior to welding of the body, turret, digger, and components shall be provided. The surface of parts to be welded shall be free of rust, scale, paint, grease, and other foreign matter. Welding rods shall be compatible with the metal being welded. All critical arc welds shall be magnaflux or dye penetrant inspected. The following defects shall not be acceptable:

(a) Welds which are ground flush with metal
(b) Slug or plug welding on load bearing members
(c) Undersize fillet welds, overlaps, inadequate fusion and penetration, undercutting of weldments and vertical down welding
(d) Unremoved slag and spatter
(e) Cracks
(f) Spot, tack, or intermittent welds for strength.

Weld penetration shall be such as to provide transference of maximum design stress through the base metal juncture. Intermittent welding shall be permissible when in accordance with 3.22.3 and 3.22.4.

3.21.2 Workmanship specimens. Workmanship specimens shall be prepared by the manufacturer of the subframe and derrick components to simulate all welding conditions. The specimens shall represent the minimum acceptable weld quality and cleaning procedure to be used in production. Upon submittal of these specimens by the contractor, the quality level and identification shall be subject to approval by the Government. The specimens shall be prepared using the welding procedures to be employed in production. Specimens prepared to represent multiple pass welds shall be made in such a manner as to leave at least 2 inches of each layer of weld metal exposed. Approval specimens shall be retained at the contractor's plant.

3.21.3 Welding procedure. The welding procedure shall be recorded prior to production of any weldment, but no procedure qualification other than workmanship specimens shall be required.

3.21.4 Factors. The factors listed below, which are applicable, shall be included in the recorded welding procedures:

(a) Joint sketch
(b) Joint type and preparation
(c) Base metal type
(d) Filler metal, size and type
(e) Position of welding

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(f) Arc-voltage, current range and polarity
(g) Manual or machine
(h) Travel speed, machine only
(i) Pre-heat and post-heat.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements (examination and tests) as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility for ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations is an acceptable practice to ascertain conformance to requirements, however this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observation of the operations to determine that practices, methods, and procedures of the contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to specification criteria has been demonstrated.

4.3 First vehicle preliminary welding inspection.

4.3.1 Welding procedure and workmanship specimen evaluation. Prior to the start of the fabrication of the subframe, body, turret, outriggers, and derrick assemblies, an inspection shall be conducted by the Government to evaluate workmanship specimens and welding procedures.

4.3.1.1 Macro-etching. Workmanship specimens shall be cross-sectioned and macro-etched. Both parts of the specimens shall be made available to the Government representatives for examination.

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4.4 First production vehicle inspection. The first vehicle produced under the contract, prior to prime painting, shall be inspected by the contractor at his plant under the direction of and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle conformance to the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the provisions of the contract.

4.4.1 Weight. The first production vehicle shall be weighed to determine the curb weight and distribution of the curb weight on front and rear axles. The total imposed loads on front and rear axles shall be computed by the contractor and verified by the Government, using the curb weight, the operator weight at 175 pounds, and the payload to provide the specified GVW or a payload of 2,500 pounds, whichever requires the greater GVW. The calculated imposed loads on the front and rear axles shall be compared to the suspension, axle and tire load capacity ratings to determine if these components are of adequate capacity to meet contractual requirements.

* 4.4.2 Road test. The vehicle, for 4x2 and 6x4 models and for models built to all wheel drive while on the chassis manufacturer's production line shall be road tested by the contractor with payload. The road test shall be for not less than 10 miles at speeds up to 52 mph. The vehicle, for all wheel drive models qualifying under 3.1.1.13, shall be road tested with and without payload. Payload shall be distributed first, so that the front axle is loaded to GAWR and the vehicle is loaded to GVW and second, with the rear axle loaded to GAWR and the vehicle loaded to GVW. The road test for each of the three conditions shall be for not less than 30 miles at speeds up to 52 mph over highways and gravel roads and for not less than 5 miles at speeds up to 15 mph over cross country terrain with ground and grade requiring all wheel drive. During the loaded and empty portions of the road test, the brakes shall be applied firmly, bringing the vehicle to a sudden stop not less than 5 times during each portion of the road test. During the road test, the front and rear suspension and the drivetrain shall be periodically inspected for interference and contact with other vehicle components. Abnormal contact of the drivetrain or suspension components with other components shall be cause for rejection. Front spring bumpers shall not make contact with frame stops except under the most extreme cross country conditions. Operational or mechanical failures of the vehicle components during the road test shall be cause for rejection. Failure includes permanent deformation as well as breakage.

* 4.4.2.1 Alternate fuel test. When operational capability on JP-4, JP-5 and JP-8 is specified, the first production vehicle shall be tested with these fuels. The contractor shall obtain a sufficient amount of JP-4, JP-5 and JP-8 to allow the first production vehicle to perform the following tests. When changing fuel types, the fuel tank and filters shall be drained, the new fuel substituted, and sufficient idling or driving shall be performed to purge the previous fuel type

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from the system. Using JP-4, the vehicle shall be warmed up for 15 minutes. Typical derrick operations shall then be performed for 15 minutes followed by 15 minutes of paved highway operation at speeds up to 50 mph. The test shall be repeated with JP-5. Using JP-8, the vehicle shall be warmed up for 15 minutes. The derrick operations shall again be performed, but for a 30 minute period, followed by a road test on paved highways at speeds up to at least 50 mph for a minimum distance of 50 miles.

4.4.3 Height, reach, and load and rotational speed tests. The vehicle shall be placed in a clear, level area, providing for unrestricted movement of the derrick. A tape measure or other measuring device shall be attached at the centerline of the upper head sheave and the derrick placed in each of the positions specified in table III. The sheave height and the angle of the derrick with the horizontal shall be measured to determine conformance to 3.3.3. During the tests, the derrick assembly shall be subject to, but not limited to, the following checks:

(a) Check to determine whether the derrick assembly can be elevated without outriggers being extended
(b) Determine if outriggers are locked automatically when in an extended position and in a retracted position
(c) Operate equipment using each set of controls separately
(d) Check to ensure that the entire system can be operated smoothly and without jerk motions in all positions
(e) Check the hydraulic system, including the pump, at 10 percent above the pump manufacturer's maximum operating pressure setting for normal workload conditions.

4.4.3.1 Load tests. The derrick shall be extended and with a measured load of not less than 1,800 pounds for size 55 and 2,200 pounds for size 70 suspended from the derrick winch line, the derrick shall be rotated, retracted, extended, elevated, lowered, stopped, restarted and the winch line raised and lowered, all in a random manner and in random combinations, at full speed, through all positions the derrick is capable of assuming. One timed revolution at maximum speed shall be made. The load shall be changed to 3,400 pounds and the tests repeated, except the derrick shall not be lowered to less than 45 degrees above horizontal. The unit shall then be tested at an angle of 75 degrees (and above, if obtainable) with the loads and with the telescoping section position restrictions specified in table III. Stalling of the vehicle; inability to rotate, elevate, lower, extend, or retract into required positions at required speeds; lifting of any vehicle wheel off the ground; part failure; or permanent set shall constitute failure of the test.

4.4.3.2 Tests with fiberglass extension. When a fiberglass extension is furnished, measurements from the sheave on the fiberglass extension and from the bottom of the aerial baskets, both to ground, shall be made to determine conformance of the extension and baskets to 3.3.3 and 3.3.4. With the derrick extended, the fiberglass extension extended, and twice the load

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specified in 3.7.3 placed in the baskets, the derrick shall be operated through all attainable positions. With the engine shut off, the baskets shall be lowered from maximum reach position to not more than 32 inches for size 55 and 40 inches for size 70 above ground level. The test shall be repeated, without baskets, with 1,350 pounds for size 55 and 1,700 pounds for size 70 suspended over the fiberglass extension head sheave. The unit shall then be tested with loads of 2,000 pounds for size 55 and 2,500 pounds for size 70, except the derrick shall not be lowered to less than 45 degrees above horizontal. The unit shall then be tested at an angle of 75 degrees (and above, if attainable) with the loads and with the telescoping section position restrictions specified in table III. Stalling of the vehicle; inability to rotate, elevate, lower, extend or retract into required positions; failure of the baskets to stay level; failure to lower at a safe rate with the engine shut off; lifting of any vehicle wheel off the ground; part failure; or permanent set shall constitute failure of the test.

4.4.3.3 Horizontal pull test. The derrick shall be placed in a horizontal position at right angles to the longitudinal centerline of the vehicle. With the derrick fully retracted and the winch line over the upper head sheave, a tension scale or dynamometer shall be placed in the winch line, parallel to the longitudinal centerline of the vehicle and attached to a fixed object. The derrick shall be rotated away from the fixed object and the winch line reeved in to take up slack and maintain the derrick at right angles to the vehicle, until the horizontal component of the force reading on the scale or dynamometer, at a right angle to the derrick, produces a torque of not less than 21,000 foot-pounds for size 55 and 37,000 foot-pounds for size 70. The horizontal component of the force is the scale or dynamometer reading times the cosine of the angle between the winch line and the horizontal. Stalling of the vehicle; malfunction; evidence of part failure or permanent set; operation of the turret hydraulic relief valve prior to reaching the required torques; or failure of the hydraulic relief valve to operate prior to overload shall constitute failure of the test.

4.4.3.4 Static tests. The vehicle shall be secured to the ground and the winch line rigged to permit operation of the derrick with overloads. Using a tension scale or dynamometer and a ground anchor, or measured weights, the derrick shall be subjected to twice the loads specified in 4.4.3.1 and 4.4.3.2 (without baskets). The derrick shall be extended and placed in its most vertical position. A load of 13,000 pounds for size 55 and 16,000 pounds for size 70 shall be suspended from the upper head sheave. Part failure or permanent set shall constitute failure of the test.

4.4.3.5 Detection of failure. After completion of the tests specified in 4.4.2 through 4.4.4.4, dye penetrant or magnaflux shall be applied on the critical areas of welds. A water-soluble penetrant shall be applied on all critical plastic areas. The water-soluble penetrant or magnaflux shall provide a means to inspect for crack indications under dynamic and static

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conditions. Any evidence of cracks or permanent deformation shall be cause for rejection of the vehicle. At the manufacturer's option, dye penetrant and magnaflux inspection may be conducted prior to the first production vehicle inspection provided all tests specified in 4.4.2 through 4.4.3.4 are conducted prior to the inspection; these tests are repeated at the first production vehicle inspection; and records of the tests and magnaflux and dye penetrant inspection witnessed by the Government are presented at the time of the first production vehicle inspection.

4.4.3.5.1 Critical area. At least the following areas shall be inspected by dye penetrant or magnaflux:

(a) Outriggers and attaching areas
(b) Turret base, trunnion, and yoke at attachment of the derrick and struts
(c) Cylinder clevis brackets
(d) Lower sheaves, sheave supporting brackets, cable anchor and support brackets
(e) Upper sheaves, sheave supporting brackets, cable anchor, and support brackets
(f) Structural steel of upper and lower derrick
(g) Upper derrick and yoke at attachment to basket and side brackets
(h) Fiberglass extension.

4.4.3.6 Temperature stabilization test. To determine conformance to 3.12.7, the vehicle shall be tested. The vehicle shall be placed in an area suitable for digging holes in firm, undisturbed soil and prepared for digger operation. The auger used shall be of the 24-inch size for size 55 vehicles and of the 30-inch size for size 70 vehicles. The ambient temperature shall be measured and recorded and a thermocouple or thermometer inserted in the hydraulic fluid reservoir at its inlet from the oil return line or, if an oil cooler is provided, at the inlet to the oil cooler. The digger shall be suspended just above the ground and continuously rotated at not less than 75 rpm. The temperature of the hydraulic oil at the hydraulic oil return line or at the inlet to the oil cooler, as applicable, shall be observed and the digger rotated until this temperature has stabilized. Immediately after the temperature stabilizes, the digger shall be used to dig a series of holes not less than 12 feet deep. Each hole shall be dug in rapid succession, with truck movement time held to the minimum, and with spin-off time held to a minimum. For hard, frozen ground or other hard soil requiring picking for removal, not less than three holes shall be dug. For all other firm, undisturbed soil conditions, not less than five holes shall be dug. A hydraulic fluid temperature exceeding 180ºF at any time during the test shall be cause for rejection.

4.4.4 Position holding ability. The vehicle shall be placed in a longitudinal position on a grade of not less than 15 percent and not leveled by use of the outriggers. The telescoping section position, height, angle, and load producing maximum tipping moment within the conditions specified in table III shall be calculated. The derrick shall be placed under these

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conditions and rotated up and down the grade to determine conformance to 3.3.4 and 3.3.5. The derrick shall be placed transverse to the longitudinal centerline of the vehicle and the vehicle engine shut off. The position of the upper head sheave relative to a fixed point shall be noted and recorded. The position of the upper head sheave relative to the same fixed point shall be noted after an elapsed period of not less than one hour to determine conformance to 3.3.7.

4.4.5 Dielectric tests. When an insulated section and baskets are furnished, dielectric tests shall be conducted at line-to-ground voltages specified in 3.7.2 or equivalent dc voltages and as follows. Except for the basket immersion test, all tests shall be conducted with all components assembled for operational use.

(a) Secure a cable from the energized conductor bar to the following areas and apply the required voltage. The total leakage in milliamperes, if any, will be indicated on the test unit and recorded.
    (1) Basket bottom, through the insulated derrick to ground, with and without basket liner
    (2) Basket front, through the insulated derrick to ground, with basket liner
    (3) Basket attachment, through the derrick to ground, with and without basket liner.
(b) Disconnect the ground cable and attach to the basket to test as follows:
Secure a cable from the energized conductor bar to the basket front and test. Secure the cable to the basket bottom and apply the required voltage, with and without basket liner. The total leakage in milliamperes indicated on the test unit shall be recorded for each test.
(c) Basket immersion test, with the basket immersed in a brine solution, and the interior of the basket filled to a depth that is practical to prevent arcing.

4.4.6 Truck body treatment and painting. A certification regarding the body cleaning, treating, prime painting and salt spray resistance testing required by MIL-STD-1223, shall be made to the Government representatives at the first production vehicle inspection.

4.4.7 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample, and shall be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

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4.4.8 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under contract until corrective action has been taken. In addition, subsequent vehicles that do not conform to the accepted production sample vehicle shall be rejected until corrective action has been taken. In the event of a failure, any or all previously accomplished tests may require repetition at the discretion of the Government.

4.4.9 Refurbishing of first production vehicle. Following successful completion of all testing, the first production vehicle shall be refurbished to like-new condition. As a minimum, refurbishing shall include repair or replacement of any damaged or severely worn parts; any necessary repainting or touch-up; cleanup of dirt, lubricants, hydraulic fluid and fuel from exterior surfaces; and any additional servicing and adjusting necessary.

4.5 Inspection of production vehicles. The contractor's inspection system shall, as a minimum, assure that the vehicle conforms to the physical and dimensional requirements and is capable of meeting performance requirements contained herein. For each vehicle under contract, the contractor shall make available to the Government, at the point of final acceptance, records acceptable to the Government indicating that the servicing and adjusting required by 3.21 have been accomplished.

5. PACKAGING

5.1 Vehicle processing. The vehicle shall be processed for shipment, from the manufacturer's plant to the initial receiving activity, in accordance with the manufacturer's standard commercial practice.

6. NOTES

6.1 Intended use. The vehicles covered by this specification are intended for general nontactical use by the Government in the performance of the maintenance and construction tasks indicated.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, model, class and size required (see 1.2).
(c) Issues of DODISS to be cited in the solicitation and if required, the specified issue of individual documents referenced (see 2.1.1 and 2.2)
(d) Identification of appropriate service for marking (see 3.1.1.2).
(e) Concealed markings, if required (see 3.1.1.2).
(f) Rustproofing, if required (see 3.1.1.3).
(g) Tropical rustproofing, if required (see 3.1.1.3).
(h) Electric trailer brake controller, if required (see 3.1.1.7).
(i) Satisfactory operation on JP-4, JP-5 and JP-8 fuels, if required (see 3.4.1.1).

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(j) Gasoline engine for class A vehicle, if required (see 3.4.1.2).
(k) Silicone rubber hoses, if required (see 3.4.1.8).
(l) Power plant heaters, if required (see 3.4.1.9).
(m) If a battery heater is not required with power plant heaters (see 3.4.1.9).
(n) A fuel fired engine preheater, if required for a diesel engine driven vehicle (see 3.4.1.10).
(o) Automatic or semiautomatic transmission, if required (see 3.4.5.5).
(p) Oil lubricated wheel bearings and axle spindles, if required (see 3.4.9).
(q) Two-speed axle, if required for model 4x2 or model 6x4 vehicle (see 3.4.9.1).
(r) Traction control, if required (see 3.4.9.3).
(s) Wide base type tires and wheels, if required (see 3.4.10).
(t) Disc type wheels, if required (see 3.4.10).
(u) Bias ply tires, if required (see 3.4.10.1).
(v) A carrier for a spare tire assembly, if required (see 3.4.10.3).
(w) Spare wheel or rim, if required (see 3.4.10.4).
(x) Spare tire assembly, if required (see 3.4.10.5).
(y) Trailer brake control system, if required (see 3.4.11.2).
(z) Brake controls for use from a towing vehicle, if required (see 3.4.11.3).
(aa) Tilting hood and fender assembly, if required (see 3.4.12).
(ab) Tilt type cab, if required for type II and type III vehicles (see 3.4.12).
(ac) Individual adjustable driver and individual passenger seats, if required (see 3.4.12.1).
(ad) Air ride driver's seat, if required (see 3.4.12.1).
(ae) Tools, if required (see 3.4.16.1).
(af) Air operated horn, in addition, if required (see 3.4.20).
(ag) Front mounted winch, if required (see 3.4.22).
(ah) Crew compartment, if required for type II vehicles (see 3.5.4).
(ai) Water cask, if required (see 3.5.6).
(aj) Hydraulic driven generator, if required (see 3.5.8).
(ak) If derrick extension and aerial basket are not required (see 3.7).
(al) Two one-man aerial baskets, if required (see 3.7.3).
(am) Augers, if required (see 3.8.1).
(an) Wire rope for derrick mounted winch in lieu of polyester, if required (see 3.9.2.1).
(ao) Dual side controls for type I, if required (see 3.10.2).
(ap) Hydraulic fluid conforming to MIL-H-83282, if required (see 3.12.5).
(aq) Hydraulic tool outlet on streetside in addition to the curbside outlet, if required (see 3.13).
(ar) Hydraulic capstan hoist, if required (see 3.13.4).
(as) Hydraulic sump pump, if required (see 3.13.5).
(at) If a basket hydraulic tool outlet is not required (see 3.14).
(au) Hydraulic chain saw, if required (see 3.14.2).

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(av) Hydraulic chain saw, pole mounted, if required (see 3.14.3).
(aw) Hydraulic circular saw, pole mounted, if required (see 3.14.4).
(ax) Hydraulic impact wrench and drill, if required (see 3.14.5).
(ay) Hydraulic pruner, pole mounted, if required (see 3.14.6).
(az) Auxiliary engine driven air compressor, if required (see 3.15).
(ba) Auxiliary engine fuel tank, if required (see 3.15.1.7).
(bb) Rear winch, if required (see 3.16).

6.3 Performance prediction. SAE Truck Ability Prediction Procedure computations and computations for low speed and maximum geared speed will be required by the contract. The SAE Work Sheet Item 1 should include vehicle model number, engine model number and vehicle type, model, class and size. Unless other conditions are cited in the contract, computations should be made for normal atmospheric pressure, normal ambient air temperature and still, dry air. The factors to be used in predicting truck ability (see 3.3.1.1) are established as follows for the corresponding SAE Truck Ability Prediction Procedure Tables:

Table 1	-	Tire Factor. This factor must relate to the size of tires furnished by the contractor in accordance with this specification.
Table 2	-	Altitude Factor. 1.00
Table 3	-	Rolling Factor. 1.613
Table 4	-	Area Factor. 0.173
Table 5	-	Velocity Factor. 250.0
Table 6	-	Altitude Factor. 1.00
Table 7	-	Chassis Friction Horsepower. Use applicable power unit GVW (to nearest, higher, tabulated 1,000 pounds) and the engine rpm (to nearest 100 revolutions) which is required for 50 mph geared speed. For GVW and engine speed beyond the range of this table, factors shall be extrapolated.
Table 8	-	Grade Factor. 0.75
Table 8A	-	Correction Factor. Not required.
Table 9	-	Road Factor. 0.0.

6.4 Instructions. The requesting service should include provisions for contractor to supply physical instructions to the using service on the proper operations of the equipment, on testing overloads, and on setting the relief valves for rated capacity.

6.5 Subject term (keyword) listing.

Aerial basket	Telescopic hydraulic derrick
Derrick, rear center	Truck, commercial
Derrick, rear corner	Truck, nontactical
Derrick, rear of cab	

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6.6 Identification of changes. The margins of this specification are marked with asterisks (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content regardless of the marginal notations and relationship to the last previous issue.

Custodians:
Army - AT
Navy - YD
Air Force - 99
Review activities:
Navy - YD
Air Force - 84

Preparing activity:
Army - AT

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